DIFFERENTIAL EQUATIONS

SEPARATION OF VARIABLES

Question 1 (**)

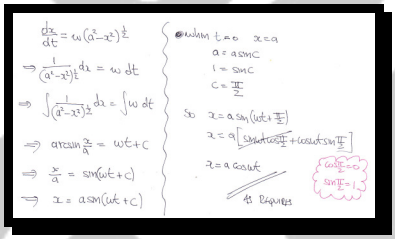
Show that if $y = a$ at $t = 0$, the solution of the differential equation

$$\frac{dy}{dt} = \omega\left(a^2 - y^2\right)^{\frac{1}{2}},$$

where a and ω are positive constants, can be written as

$$y = a\cos\omega t\,.$$

proof



Question 2 (+)**

Show that a general solution of the differential equation

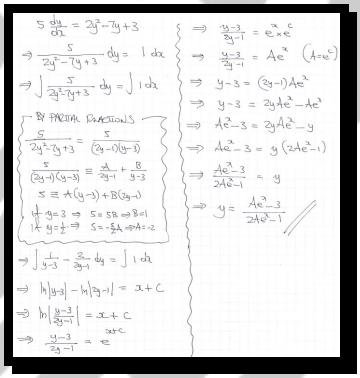
$$5\frac{dy}{dx}=2y^2-7y+3$$

is given by

$$y=\frac{A\mathrm{e}^x-3}{2A\mathrm{e}^x-1},$$

where A is an arbitrary constant.

proof



Question 3 (+)**

Show that a general solution of the differential equation

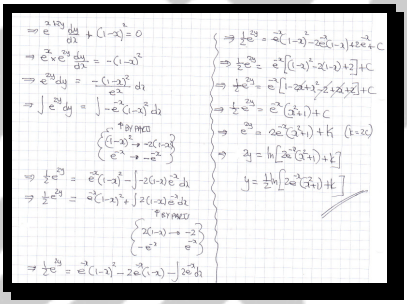
$$e^{x+2y}\frac{dy}{dx}+(1-x)^2=0$$

is given by

$$y=\frac{1}{2}\ln\left[2e^{-x}\left(x^2+1\right)+K\right],$$

where K is an arbitrary constant.

proof



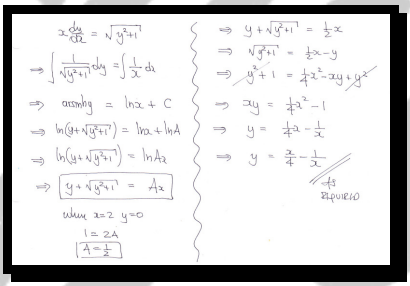
Question 4 (+)**

$$x\frac{dy}{dx}=\sqrt{y^2+1}\ ,\ x>0\text{, with }\ y=0\ \text{ at }\ x=2\,.$$

Show that the solution of the above differential equation is

$$y=\frac{x}{4}-\frac{1}{x}\,.$$

proof



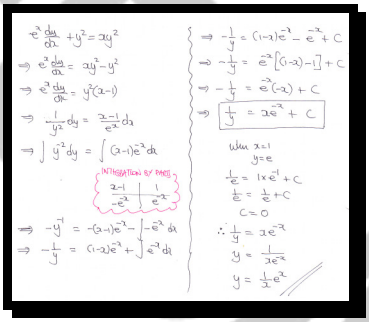
Question 5 (***)

$$\mathrm{e}^x\frac{dy}{dx}+y^2=xy^2\,,\ x>0\,,\ y>0$$

Show that the solution of the above differential equation subject to $y=\mathrm{e}$ at $x=1$, is

$$y=\frac{1}{x}\mathrm{e}^x\,.$$

proof



Question 6 (***)

A curve $y=f(x)$ satisfies the differential equation

$$y=1-\frac{dy}{dx}\frac{x+1}{(x-1)(x+2)},\ y>1, x>-1$$

a) Solve the differential equation to show that

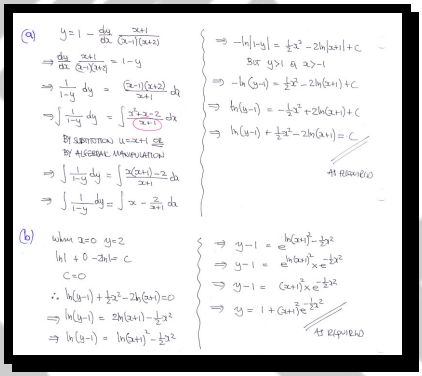
$$\ln(y-5)+\frac{1}{2}x^2+4x-2\ln(x+1)=C.$$

When $x=0$, $y=2$.

b) Show further that

$$y=1+(x+1)^2\,\mathrm{e}^{-\frac{1}{2}x^2}.$$

proof



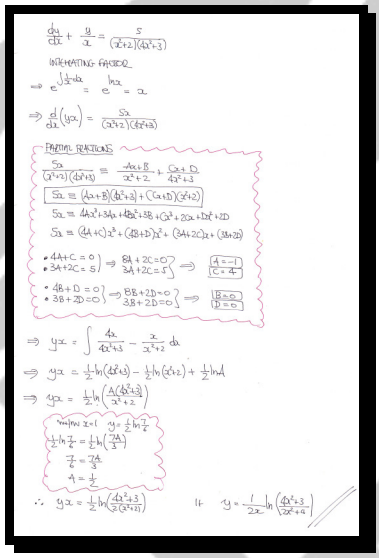
Question 7 (***)

$$\frac{dy}{dx}+\frac{y}{x}=\frac{5}{\left(x^2+2\right)\left(4x^2+3\right)},\ x>0.$$

Given that $y=\frac{1}{2}\ln\frac{7}{6}$ at $x=1$, show that the solution of the above differential equation can be written as

$$y=\frac{1}{2x}\ln\left(\frac{4x^2+3}{2x^2+4}\right).$$

proof

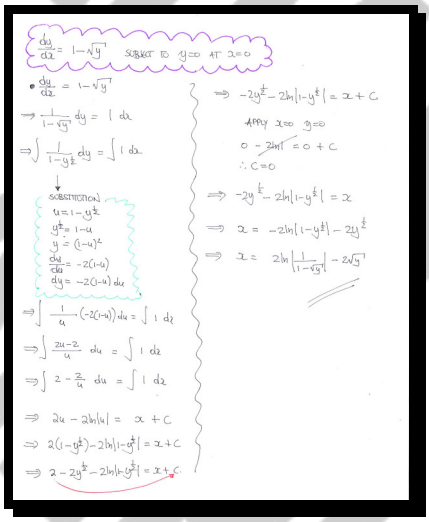


Question 8 (***)

$$\frac{dy}{dx}=1-\sqrt{y}\ ,\ \ y\geq 0,\ \ y\neq 1\ .$$

Find the solution of the above differential equation subject to the condition $y=0$ at $x=0$, giving the answer in the form $x=f\left(y\right)$.

$$\boxed{x=2\ln\left|\frac{1}{1-\sqrt{y}}\right|-2\sqrt{y}}$$



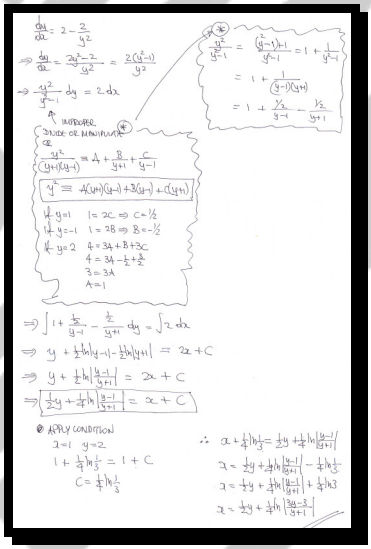
Question 9 (***)

Solve the differential equation

$$\frac{dy}{dx}=2-\frac{2}{y^2},$$

subject to the condition $y=2$ at $x=1$, giving the answer in the form $x=f\left(y\right)$.

$$\boxed{x=\frac{1}{2}y+\frac{1}{4}\ln\left|\frac{3y-3}{y+1}\right|}$$



Question 10 (*+)**

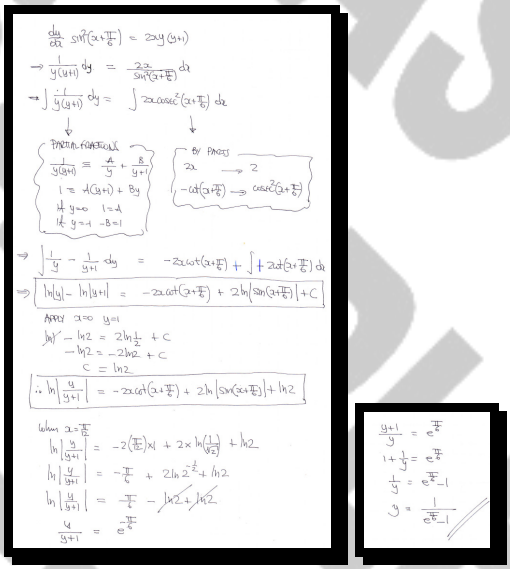
The function $y = f(x)$ satisfies the differential equation

$$\frac{dy}{dx}\sin^2\left(x+\frac{\pi}{6}\right) = 2xy(y+1),$$

subject to the condition $y = 1$ at $x = 0$.

Find the exact value of y when $x = \frac{\pi}{12}$.

$$\boxed{y = \frac{1}{e^{\frac{\pi}{6}} - 1}}$$



1ST ORDER BY STANDARD INTEGRATING FACTORS

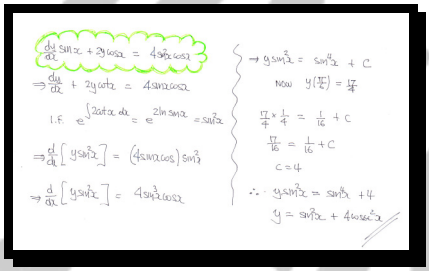
Question 1 (**)

Solve the differential equation

$$\frac{dy}{dx}\sin x+2y\cos x=4\sin^2 x\cos x\,,\qquad y\left(\tfrac{1}{6}\pi\right)=\tfrac{17}{4}\,.$$

Give the answer in the form $y=f\left(x\right)$.

$$\boxed{y=\sin^2 x+4\operatorname{cosec}^2 x}$$

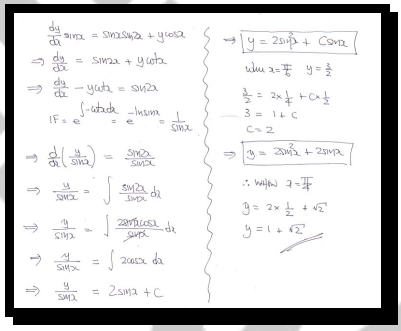


Question 2 (**)

$$\frac{dy}{dx}\sin x=\sin x\sin 2x+y\cos x\,.$$

Given that $y=\dfrac{3}{2}$ at $x=\dfrac{\pi}{6}$, find the exact value of y at $x=\dfrac{\pi}{4}$.

$$\boxed{1+\sqrt{2}}$$



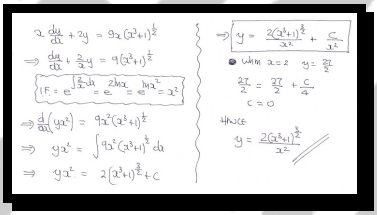
Question 3 (**)

$$x\frac{dy}{dx}+2y=9x\left(x^3+1\right)^{\frac{1}{2}} \text{ , with } y=\frac{27}{2} \text{ at } x=2 .$$

Show that the solution of the above differential equation is

$$y=\frac{2}{x^2}\left(x^3+1\right)^{\frac{3}{2}} .$$

proof



Question 4 (**)

20 grams of salt are dissolved into a beaker containing 1 litre of a certain chemical.

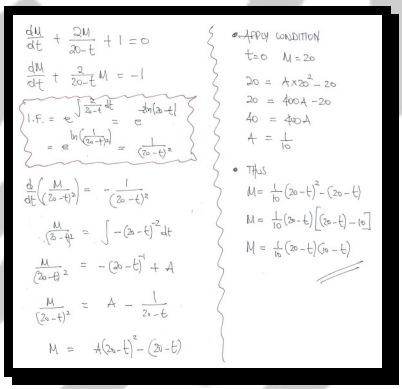
The mass of salt, M grams, which remains undissolved t seconds later, is modelled by the differential equation

$$\frac{dM}{dt}+\frac{2M}{20-t}+1=0\,,\ t\geq 0\,.$$

Show clearly that

$$M=\frac{1}{10}(10-t)(20-t)\,.$$

proof

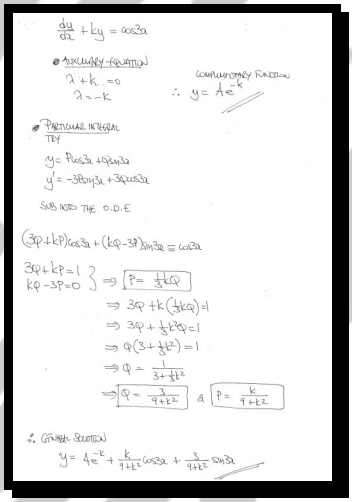


Question 5 (+)**

$$\frac{dy}{dx}+ky=\cos 3x\,,\ k \text{ is a non zero constant.}$$

By finding a complimentary function and a particular integral, or otherwise, find the general of the above differential equation.

$$\boxed{y=A\mathrm{e}^{-x}+\frac{k}{9+k^2}\cos 3x+\frac{3}{9+k^2}\sin 3x}$$



Question 6 (+)**

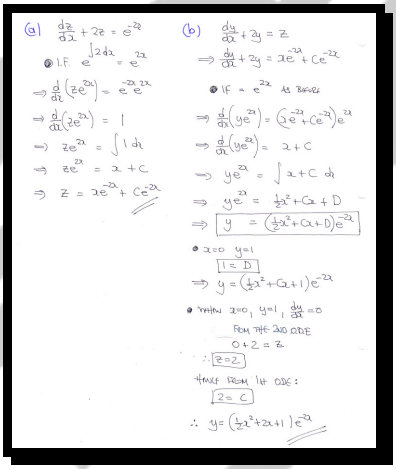
Given that $z = f(x)$ and $y = g(x)$ satisfy the following differential equations

$$\frac{dz}{dx} + 2z = \mathrm{e}^{-2x} \quad \text{and} \quad \frac{dy}{dx} + 2y = z,$$

a) Find z in the form $z = f(x)$

b) Express y in the form $y = g(x)$, given further that at $x = 0$, $y = 1$, $\frac{dy}{dx} = 0$

$$\boxed{z = (x + C)\mathrm{e}^{-2x}}, \quad \boxed{y = \left(\frac{1}{2}x^2 + 2x + 1\right)\mathrm{e}^{-2x}}$$



Question 7 (***)

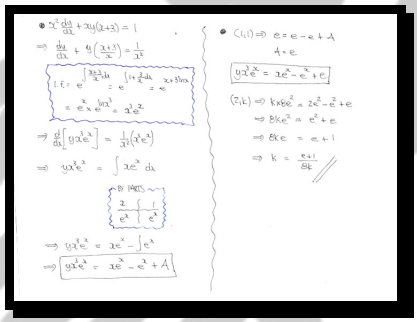
A curve C, with equation $y = f(x)$, passes through the points with coordinates $(1,1)$ and $(2,k)$, where k is a constant.

Given further that the equation of C satisfies the differential equation

$$x^2 \frac{dy}{dx} + xy(x+3) = 1,$$

determine the exact value of k.

$$\boxed{k = \frac{e+1}{8e}}$$



Question 8 (***)

A curve C, with equation $y = f(x)$, meets the y axis the point $(0,1)$.

It is further given that the equation of C satisfies the differential equation

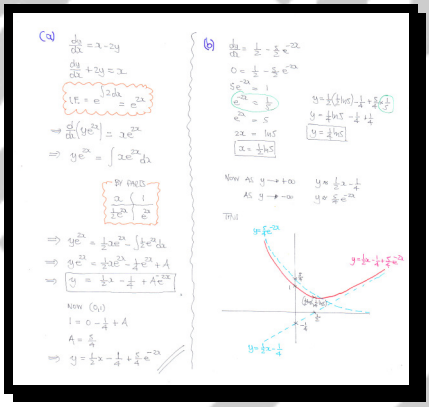
$$\frac{dy}{dx} = x - 2y .$$

a) Determine an equation of C.

b) Sketch the graph of C.

The graph must include in exact simplified form the coordinates of the stationary point of the curve and the equation of its asymptote.

$$\boxed{y = \frac{1}{2}x - \frac{1}{4} + \frac{5}{4}e^{-2x}}$$



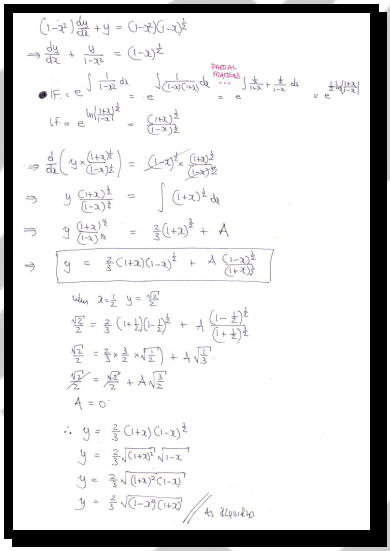
Question 9 (***)

$$\left(1-x^2\right)\frac{dy}{dx}+y=\left(1-x^2\right)(1-x)^{\frac{1}{2}}\ ,\ -1<x<1\,.$$

Given that $y=\dfrac{\sqrt{2}}{2}$ at $x=\dfrac{1}{2}$, show that the solution of the above differential equation can be written as

$$y=\frac{2}{3}\sqrt{\left(1-x^2\right)(1+x)}\,.$$

proof



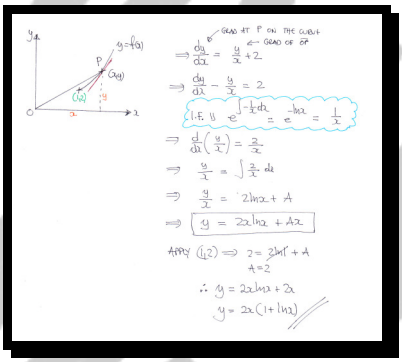
Question 10 (***)

The general point P lies on the curve with equation $y = f(x)$.

The gradient of the curve at P is 2 more than the gradient of the straight line segment OP.

Given further that the curve passes through $Q(1,2)$, express y in terms of x.

$$\boxed{y = 2x(1+\ln x)}$$



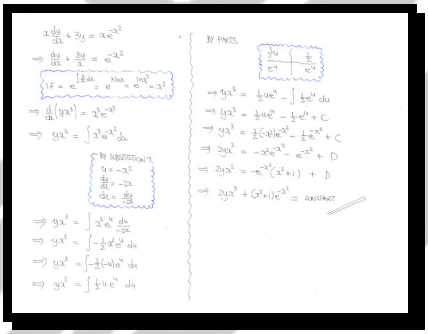
Question 11 (***)

$$x\frac{dy}{dx}+3y=x\mathrm{e}^{-x^2}\ ,\ x>0\,.$$

Show clearly that the general solution of the above differential equation can be written in the form

$$2yx^3+\left(x^2+1\right)\mathrm{e}^{-x^2}=\text{constant}\,.$$

proof



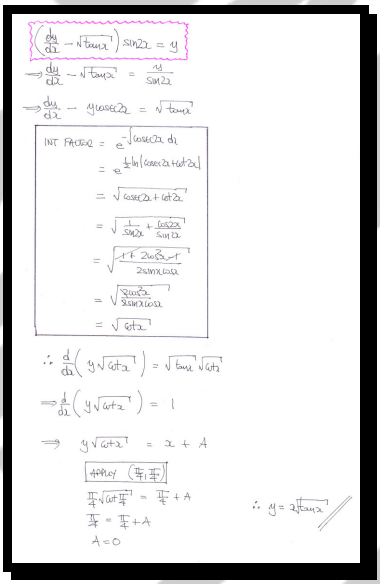
Question 12 (****)

It is given that a curve with equation $y = f(x)$ passes through the point $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$ and satisfies the differential equation

$$\left(\frac{dy}{dx} - \sqrt{\tan x}\right)\sin 2x = y .$$

Find an equation for the curve in the form $y = f(x)$.

$$\boxed{y = x\sqrt{\tan x}}$$



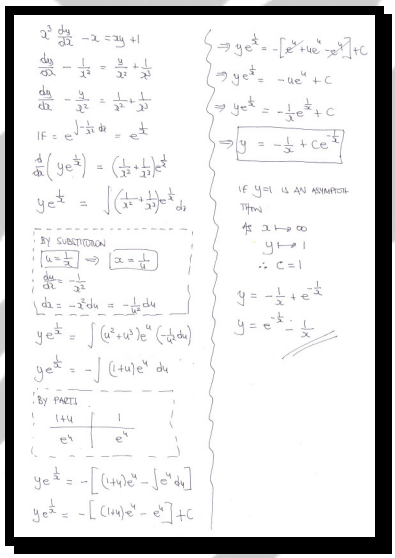
Question 13 (****)

The curve with equation $y = f(x)$ has the line $y = 1$ as an asymptote and satisfies the differential equation

$$x^3 \frac{dy}{dx} - x = xy + 1, \ x \neq 0 .$$

Solve the above differential equation, giving the solution in the form $y = f(x)$.

$$\boxed{y = e^{-\frac{1}{x}} - \frac{1}{x}}$$



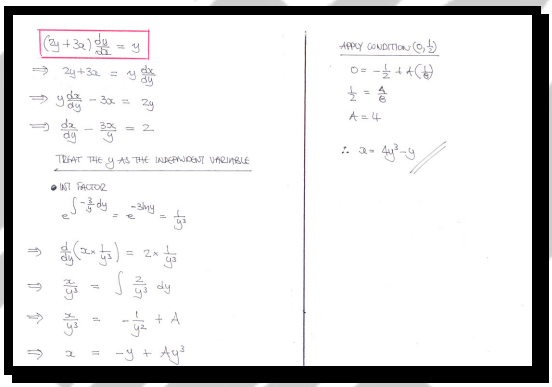
Question 24 (****)

It is given that a curve with equation $x = f(y)$ passes through the point $\left(0, \frac{1}{2}\right)$ and satisfies the differential equation

$$(2y+3x)\frac{dy}{dx} = y.$$

Find an equation for the curve in the form $x = f(y)$.

$$\boxed{x = 4y^3 - y}$$



Question 25 (****)

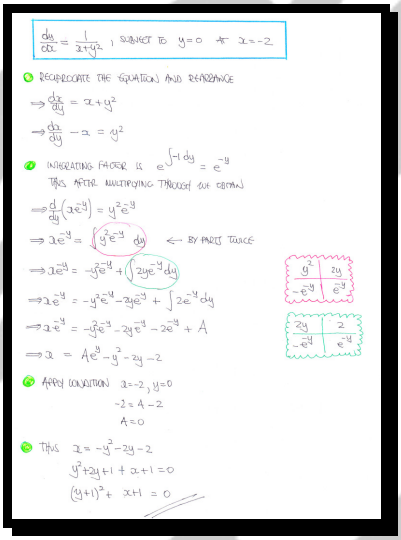
It is given that a curve passes through the point $(-2,0)$ and satisfies the ordinary differential equation

$$\frac{dy}{dx}=\frac{1}{x+y^2}.$$

Show that an equation of C is

$$(y+1)^2+x+1=0.$$

proof



1ST ORDER HOMOGENEOUS

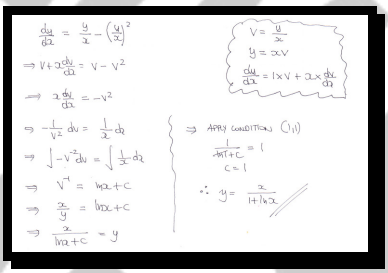
Question1 (+)**

Solve the differential equation

$$\frac{dy}{dx}=\frac{y}{x}-\left(\frac{y}{x}\right)^2,\ x>0,$$

subject to the condition $y=1$ at $x=1$.

$$\boxed{y=\frac{x}{1+\ln x}}$$



Question 2 (+)**

$$\frac{dy}{dx}=\frac{(4x+y)(x+y)}{x^2},\ x>0.$$

a) Use a suitable substitution to show that the above differential equation can be transformed to

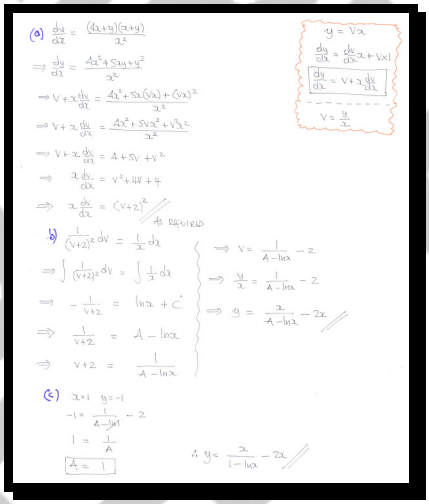
$$x\frac{dv}{dx}=(v+2)^2.$$

b) Hence find the general solution of the original differential equation, giving the answer in the form $y=f(x)$.

c) Use the boundary condition $y=-1$ at $x=1$, to show that a specific solution of the original differential equation is

$$y=\frac{x}{1-\ln x}-2x.$$

$$\boxed{y=\frac{x}{A-\ln x}-2x}$$



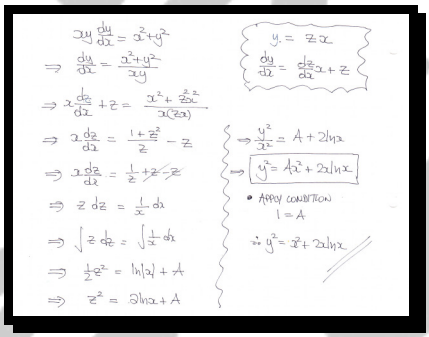
Question 3 (+)**

By using a suitable substitution, solve the differential equation

$$xy\frac{dy}{dx}=x^2+y^2,\ x>0,$$

subject to the boundary condition $y=1$ at $x=1$.

$$\boxed{y=x^2\left(1+2\ln x\right)}$$



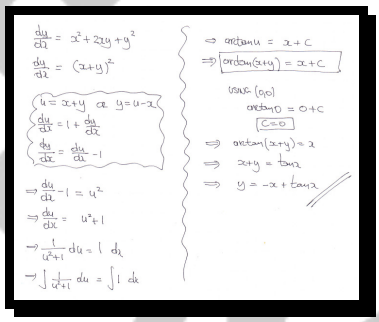
Question 4 (+)**

By using a suitable substitution, or otherwise, solve the differential equation

$$\frac{dy}{dx}=x^2+2xy+y^2,$$

subject to the condition $y(0)=0$.

$$\boxed{y=-x+\tan x}$$



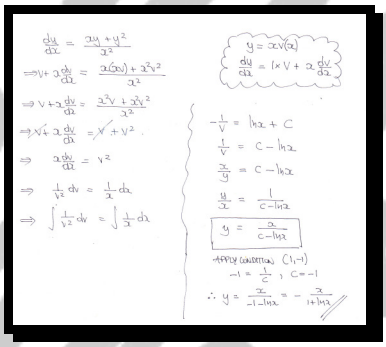
Question 5 (+)**

By using a suitable substitution, solve the differential equation

$$\frac{dy}{dx}=\frac{xy+y^2}{x^2},\ x>0,$$

subject to the condition $y=-1$ at $x=1$.

$$\boxed{y=-\frac{x}{1+\ln x}}$$



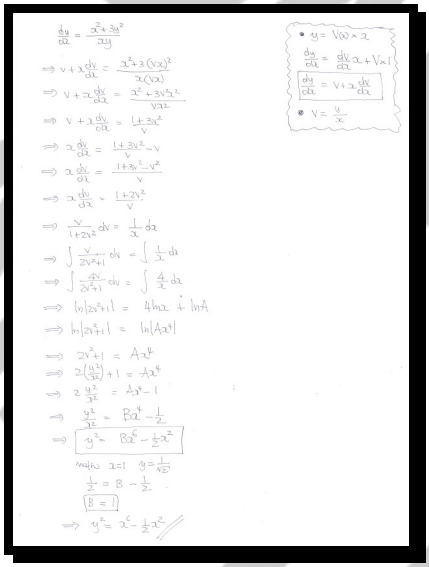
Question 6 (***)

$$\frac{dy}{dx}=\frac{x^2+3y^2}{xy},\ x>0,\ y>0.$$

Given the boundary condition $y(1)=\frac{1}{\sqrt{2}}$, show that

$$y^2=x^6-\frac{1}{2}x^2.$$

proof



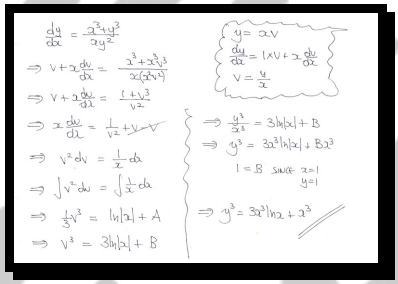
Question 7 (***)

By using a suitable substitution, solve the differential equation

$$\frac{dy}{dx}=\frac{x^3+y^3}{xy^2},$$

subject to the condition $y=1$ at $x=1$.

$$\boxed{y^3=x^3\left(3\ln x+1\right)}$$



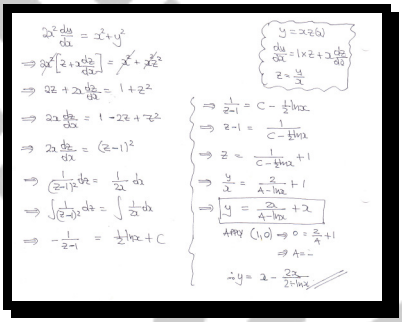
Question 8 (***)

By using a suitable substitution, solve the differential equation

$$2x^2\frac{dy}{dx}=x^2+y^2,\ x>0,$$

subject to the condition $y(1)=0$.

$$\boxed{y=x-\frac{2x}{2+\ln x}}$$



Question 9 (***)

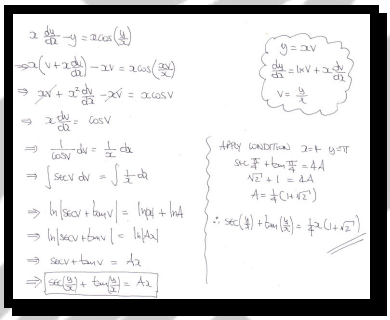
By using a suitable substitution, solve the differential equation

$$x\frac{dy}{dx}-y=x\cos\left(\frac{y}{x}\right),\ x\neq 0,$$

subject to the condition $y(4)=\pi$.

The final answer may not involve natural logarithms.

$$\boxed{\sec\left(\frac{y}{x}\right)+\tan\left(\frac{y}{x}\right)=\frac{1}{4}x\left(1+\sqrt{2}\right)}$$



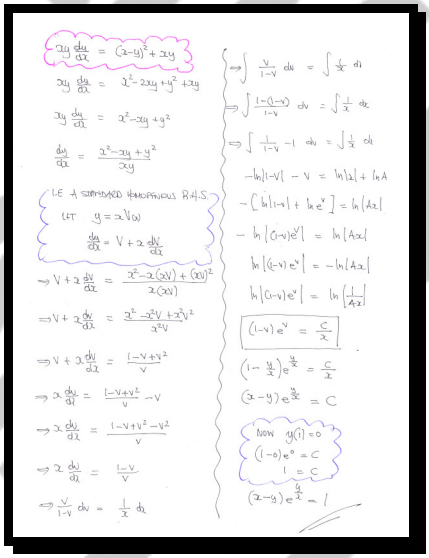
Question 10 (***)

$$xy\frac{dy}{dx}=(x-y)^2+xy\,,\quad y(1)=0\,.$$

Show that the solution of the above differential equation is

$$(x-y)\mathrm{e}^{\frac{y}{x}}=1\,.$$

proof



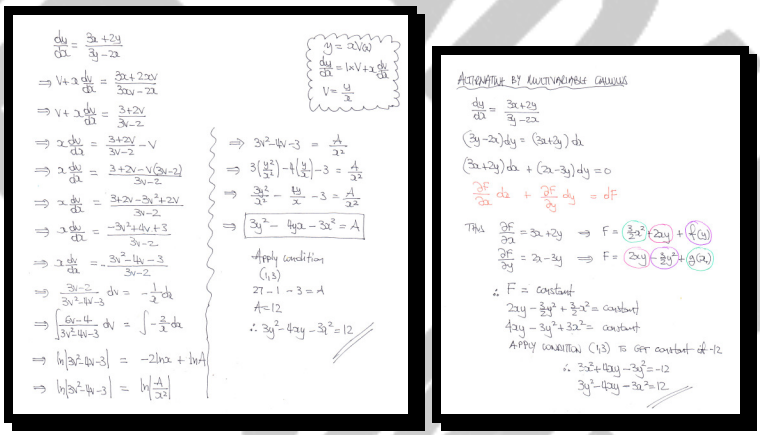
Question 11 (***)

Solve the differential equation

$$\frac{dy}{dx}=\frac{3x+2y}{3y-2x}, \quad y(1)=3.$$

subject to the condition $y=3$ at $x=1$.

$$\boxed{3y^2-4xy-3x^2=12}$$



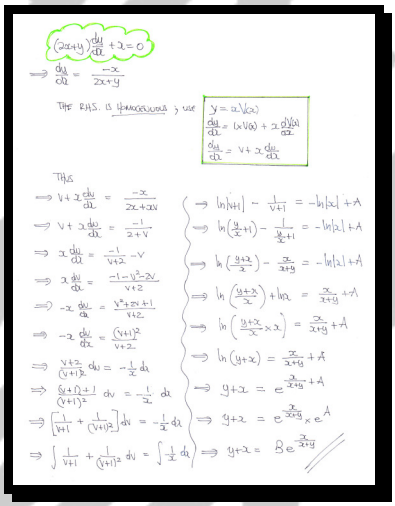
Question 12 (*+)**

Find a general solution for the following differential equation

$$(2x+y)\frac{dy}{dx}+x=0\,.$$

The final answer must not contain natural logarithms.

$$\boxed{y+x=A\,\mathrm{e}^{\frac{x}{x+y}}}$$

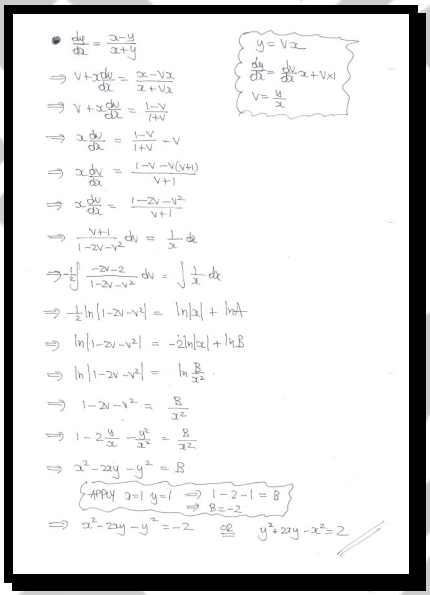


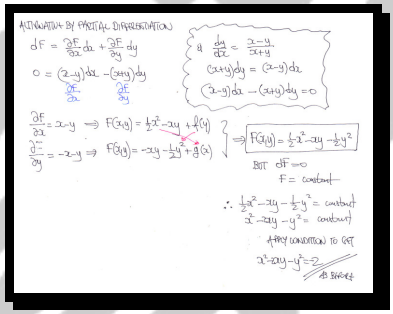
Question 13 (****)

Solve the differential equation

$$\frac{dy}{dx}=\frac{x-y}{x+y},\ y(1)=1.$$

$$\boxed{y^2+2xy-x^2=2}$$





Question 14 (****)

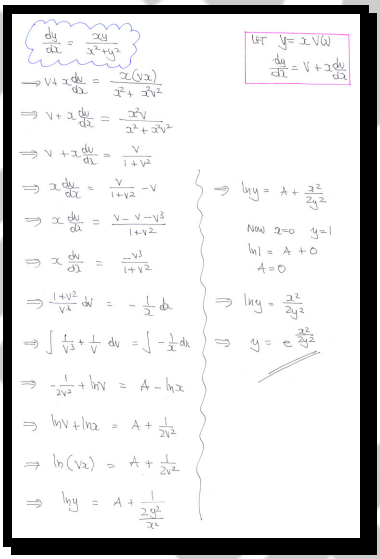
It is given that a curve with equation $f(x,y)=0$ passes through the point $(0,1)$ and satisfies the differential equation

$$\frac{dy}{dx}=\frac{xy}{x^2+y^2}.$$

By solving the differential equation, show that an equation for the curve is

$$y=\exp\left[\frac{x^2}{2y^2}\right].$$

proof



1^{ST} ORDER BERNOULLI TYPE

Question 1 (+)**

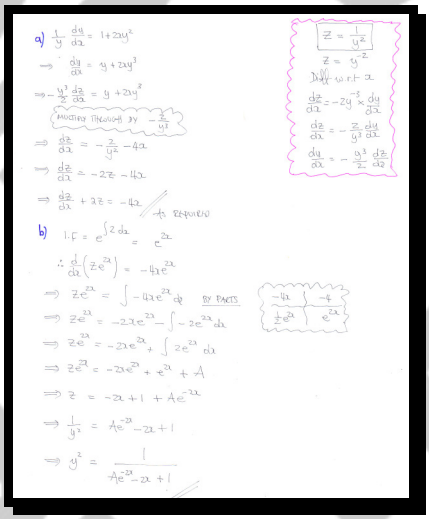
$$\frac{1}{y}\frac{dy}{dx}=1+2xy^2\,,\ \ y>0\,.$$

a) Show that the substitution $z=\dfrac{1}{y^2}$ transforms the above differential equation into the new differential equation

$$\frac{dz}{dx}+2z=-4x\,.$$

b) Hence find the general solution of the original differential equation, giving the answer in the form $y^2=f\left(x\right)$.

$$\boxed{y^2=\frac{1}{Ae^{x}-2x+1}}$$

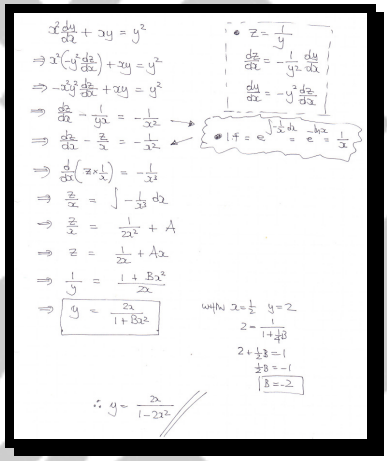


Question 2 (***)

Solve the differential equation

$$x^2\frac{dy}{dx}+xy=y^2,\ \ y\left(\frac{1}{2}\right)=2.$$

$$\boxed{y=\frac{2x}{1-2x^2}}$$

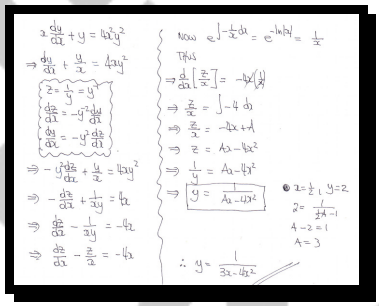


Question 3 (***)

Solve the differential equation

$$x\frac{dy}{dx}+y=4x^2y^2,\quad y\left(\frac{1}{2}\right)=2.$$

$$\boxed{y=\frac{1}{3x-4x^2}}$$



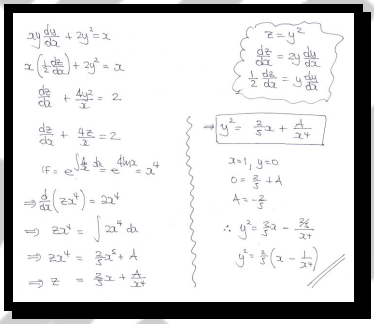
Question 4 (***)

By using a suitable substitution, solve the differential equation

$$xy\frac{dy}{dx}+2y^2=x,\ y(1)=0.$$

Give the answer in the form $y^2=f(x)$.

$$\boxed{y^2=\frac{2}{5}\left(x-\frac{1}{x^4}\right)}$$



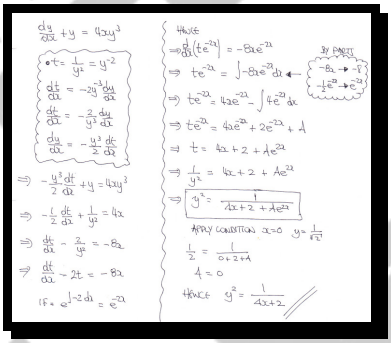
Question 5 (***)

Solve the differential equation

$$\frac{dy}{dx}+y=4xy^3,\ y(0)=\frac{1}{\sqrt{2}}.$$

Give the answer in the form $y^2=f(x)$.

$$\boxed{y^2=\frac{1}{4x+2}}$$



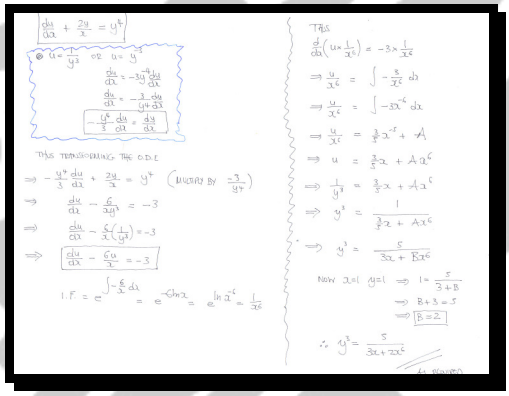
Question 6 (*+)**

$$\frac{dy}{dx}+\frac{2y}{x}=y^4,\ x>0,\ y>0.$$

Given that $y(1)=1$, show that

$$y^3=\frac{5}{3x+2x^6}.$$

proof



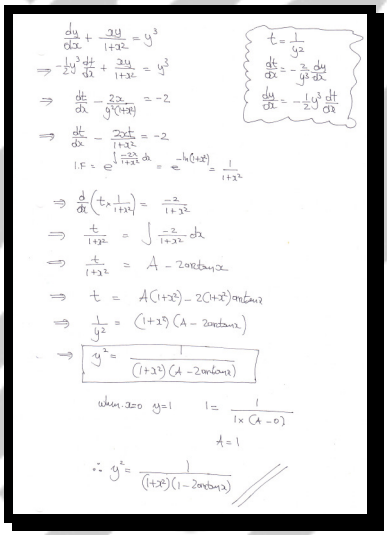
Question 7 (*+)**

Solve the differential equation

$$\frac{dy}{dx}+\frac{xy}{1+x^2}=y^3,\ y(0)=1.$$

Give the answer in the form $y^2=f(x)$.

$$\boxed{y^2=\frac{1}{\left(1+x^2\right)\left(1-2\arctan x\right)}}$$

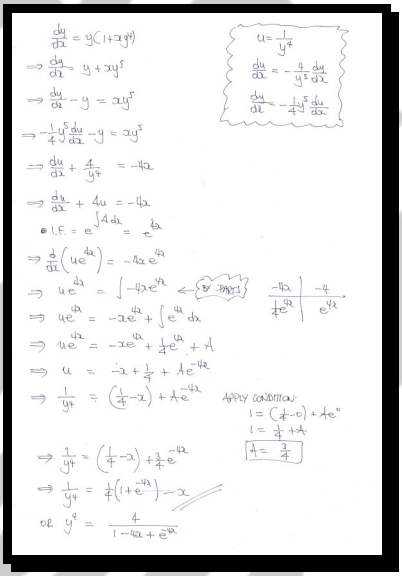


Question 8 (*+)**

Solve the differential equation

$$\frac{dy}{dx}=y\left(1+xy^4\right),\ \ y\left(0\right)=1.$$

$$\boxed{\frac{1}{y^4}=\frac{1}{4}\left(1+3\mathrm{e}^{-4x}\right)-x}$$



Question 9 (****)

A curve C passes through the point $(1,1)$ and satisfies the differential equation

$$\frac{dy}{dx}-\frac{y}{x}=\frac{x^3}{4y^3},\ x>0,\ y>0,$$

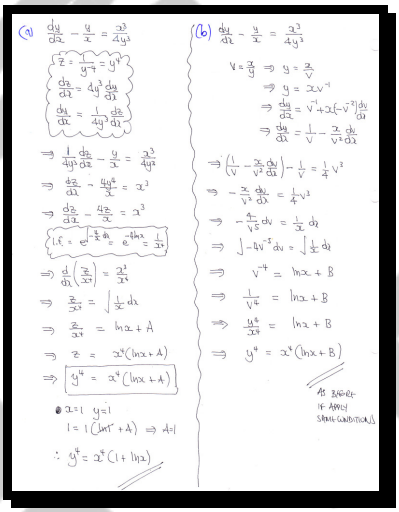
subject to the condition $y=1$ at $x=1$.

a) Find an equation of C by using the substitution $z=y^4$.

b) Find an equation of C by using the substitution $v=\frac{x}{y}$.

Give the answer in the form $y^4=f(x)$.

$$\boxed{y^4=x^4(1+\ln x)}$$



1ST ORDER BY PARTIAL DIFFERENTIATION TECHNIQUES

Question 1 (+)**

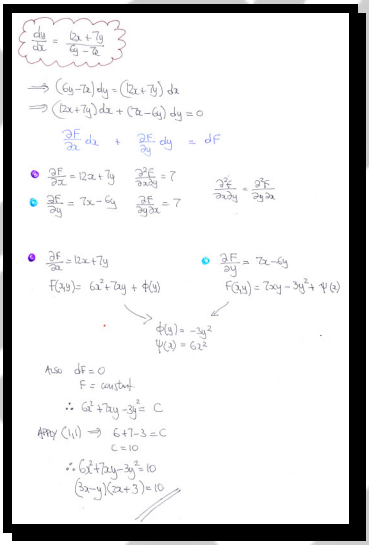
$$\frac{dy}{dx}=\frac{12x+7y}{6y-7x},\ y(1)=1.$$

Use a method involving partial differentiation to show that the solution of the above differential equation can be written as

$$(ax+by)(cx+dy)=10,$$

where a, b, c and d are integers to be found.

$$\boxed{(3x-y)(2x+3y)=k}$$



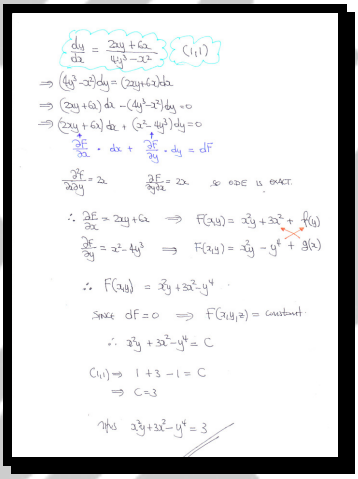
Question 2 (***)

Solve the differential equation

$$\frac{dy}{dx}=\frac{2xy+6x}{4y^3-x^2},$$

subject to the boundary condition $y=1$ at $x=1$.

$$\boxed{x^2y+3x^2-y^4=3}$$



$\frac{dy}{dx} = \frac{2xy+6x}{4y^3-x^2}$ $(1,1)$

$\Rightarrow (4y^3-x^2)dy = (2xy+6x)dx$

$\Rightarrow (2xy+6x)dx - (4y^3-x^2)dy = 0$

$\Rightarrow (2xy+6x)dx + (x^2-4y^3)dy = 0$

$\frac{\partial F}{\partial x}\cdot dx + \frac{\partial F}{\partial y}\cdot dy = dF$

$\frac{\partial^2 F}{\partial x \partial y} = 2x$ $\quad \frac{\partial^2 F}{\partial y \partial x} = 2x$ $\quad$ so ODE is EXACT

$\therefore \frac{\partial F}{\partial x} = 2xy+6x \Rightarrow F(x,y) = x^2y + 3x^2 + f(y)$

$\frac{\partial F}{\partial y} = x^2 - 4y^3 \Rightarrow F(x,y) = x^2y - y^4 + g(x)$

$\therefore F(x,y) = x^2y + 3x^2 - y^4$

since $dF = 0 \Rightarrow F(x,y,z) =$ constant

$\therefore x^2y + 3x^2 - y^4 = C$

$(1,1) \Rightarrow 1+3-1 = C$

$\Rightarrow C = 3$

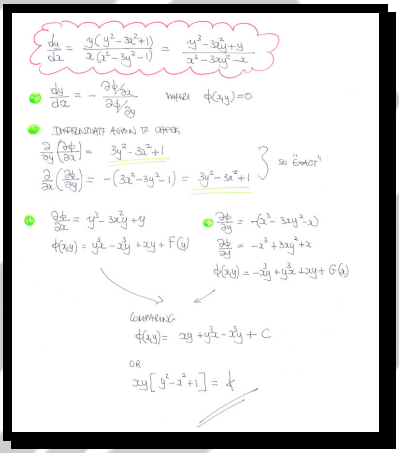
$\therefore x^2y + 3x^2 - y^4 = 3$

Question 3 (***)

Find a general solution of the following differential equation

$$\frac{dy}{dx}=\frac{y\left(y^2-3x^2+1\right)}{x\left(x^2-3y^2-1\right)}.$$

$$\boxed{xy\left(x^2-y^2-1\right)=\text{constant}}$$



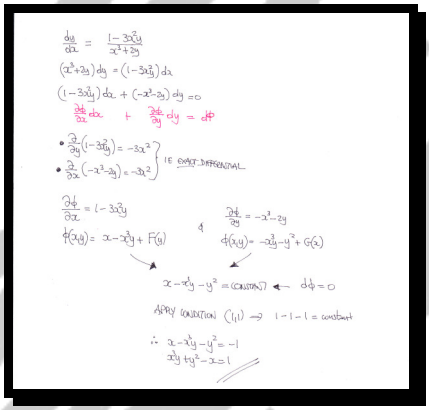
Question 4 (***)

Find the solution of the following differential equation

$$\frac{dy}{dx}=\frac{1-3x^2y}{x^3+2y},$$

subject to the boundary condition $y=1$ at $x=1$.

$$\boxed{x^3y+y^2-x=1}$$



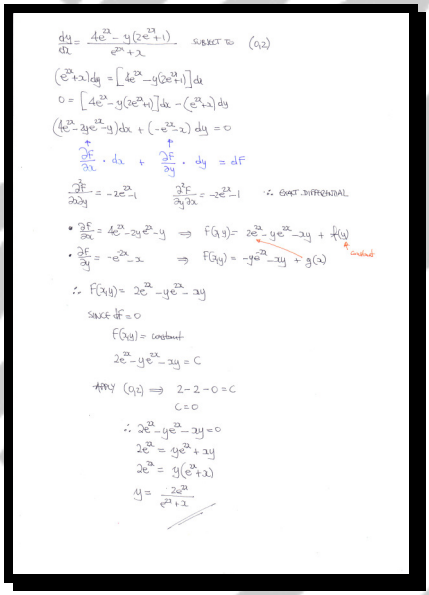
Question 5 (***)

Solve the differential equation

$$\frac{dy}{dx}=\frac{4\mathrm{e}^{2x}-y\left(2\mathrm{e}^{2x}+1\right)}{\mathrm{e}^{2x}+x},$$

subject to the boundary condition $y=2$ at $x=0$.

$$\boxed{y=\frac{2\mathrm{e}^{2x}}{\mathrm{e}^{2x}+x}}$$

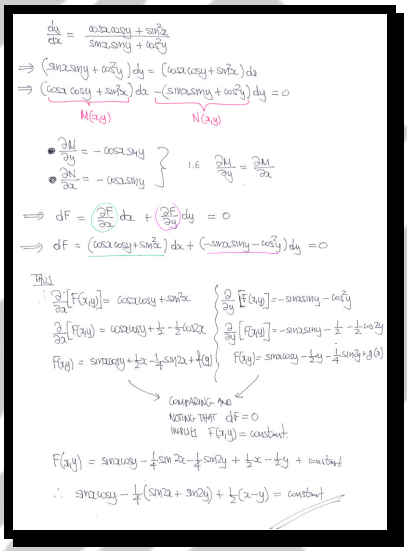


Question 6 (*+)**

Find a general solution of the following differential equation

$$\frac{dy}{dx}=\frac{\cos x\cos y+\sin^2 x}{\sin x\sin y+\cos^2 y}.$$

$$\boxed{\sin x\cos y-\frac{1}{4}\left(\sin 2x+\sin 2y\right)+\frac{1}{2}\left(x-y\right)=\text{constant}}$$

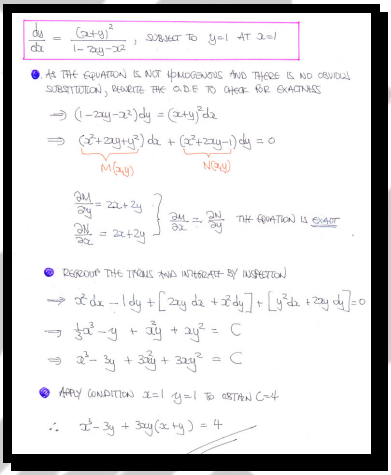


Question 7 (***)

Determine the solution of the following differential equation by looking for a suitable integrating factor.

$$\frac{dy}{dx}=\frac{(x+y)^2}{1-2xy-x^2}, \quad y(1)=1.$$

$$\boxed{x^3-3y+3xy(x+y)=4}$$

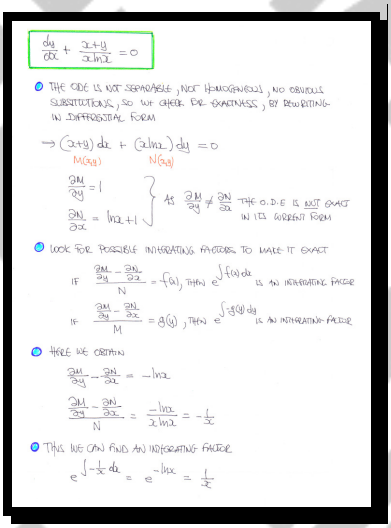


Question 8 (*+)**

Find a general solution of the following differential equation by looking for a suitable integrating factor.

$$\frac{dy}{dx}+\frac{x+y}{x\ln x}=0\,.$$

$$\boxed{x+y\ln x=C}$$



$\frac{dy}{dx}+\frac{x+y}{x\ln x}=0$

- THE ODE IS NOT SEPARABLE, NOT HOMOGENEOUS, NO OBVIOUS SUBSTITUTIONS, SO WE CHECK FOR EXACTNESS, BY REWRITING IN DIFFERENTIAL FORM

$\rightarrow (x+y)\,dx+(x\ln x)\,dy=0$

$M(x,y)$ $N(x,y)$

$\frac{\partial M}{\partial y}=1$, $\frac{\partial N}{\partial x}=\ln x+1$ } AS $\frac{\partial M}{\partial y}\neq\frac{\partial N}{\partial x}$ THE O.D.E IS NOT EXACT IN ITS CURRENT FORM

- LOOK FOR POSSIBLE INTEGRATING FACTORS TO MAKE IT EXACT

IF $\frac{\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}}{N}=f(x)$, THEN $e^{\int f(x)\,dx}$ IS AN INTEGRATING FACTOR

IF $\frac{\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}}{M}=g(y)$, THEN $e^{\int g(y)\,dy}$ IS AN INTEGRATING FACTOR

- HERE WE OBTAIN

$\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}=-\ln x$

$\frac{\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}}{N}=\frac{-\ln x}{x\ln x}=-\frac{1}{x}$

- THUS WE CAN FIND AN INTEGRATING FACTOR

$e^{\int -\frac{1}{x}\,dx}=e^{-\ln x}=\frac{1}{x}$

- MULTIPLY THE O.D.E BY $\frac{1}{x}$, AND INTEGRATE BY INSPECTION

$\Rightarrow \left(1+\frac{y}{x}\right)dx+(\ln x)\,dy=0$

$\Rightarrow 1\,dx+\left(\frac{y}{x}\,dx+\ln x\,dy\right)=0$

$\Rightarrow x+y\ln x=C$

ALTERNATIVE WITHOUT INSPECTION

$\left(1+\frac{y}{x}\right)dx+(\ln x)\,dy=0$

$\frac{\partial\phi}{\partial x}\cdot dx+\frac{\partial\phi}{\partial y}\cdot dy=d\phi$

- $\frac{\partial\phi}{\partial x}=1+\frac{y}{x}$, $\phi(x,y)=x+y\ln x+f(y)$
- $\frac{\partial\phi}{\partial y}=\ln x$, $\phi(x,y)=y\ln x+g(x)$

COMPARING: $g(x)=x$, $f(y)=$ CONSTANT.

SO $d\phi=0$, $\phi(x,y)=C$, $x+y\ln x=C$

Question 9 (*+)**

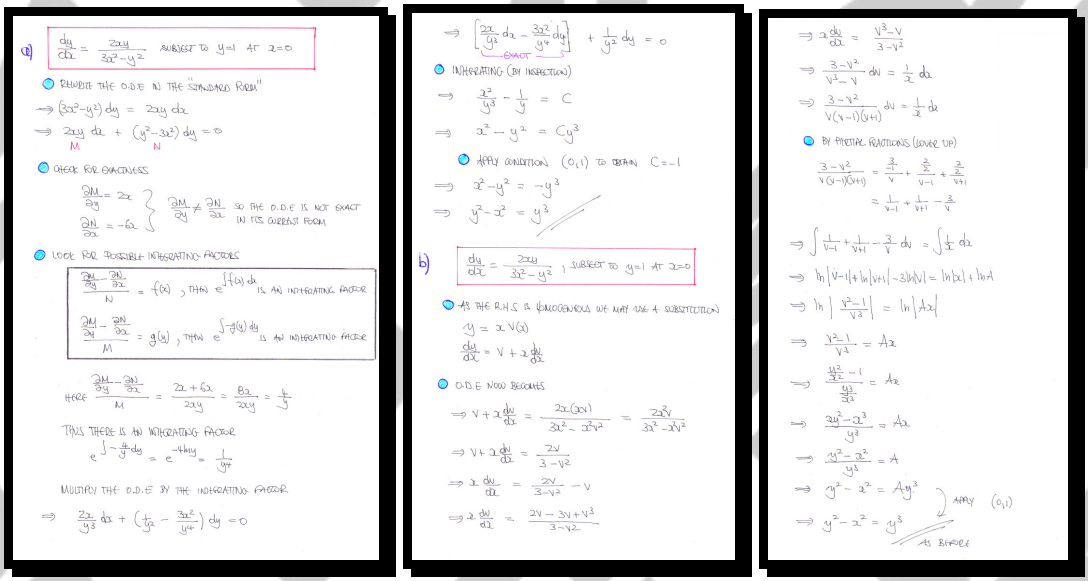
$$\frac{dy}{dx}=\frac{2xy}{3x^2-y^2},\ y(0)=1.$$

a) Find an integrating factor for the above differential equation and hence show

$$y^3=y^2-x^2.$$

b) Verify the answer of part **(a)** by a solving the differential equation by a suitable substitution.

proof

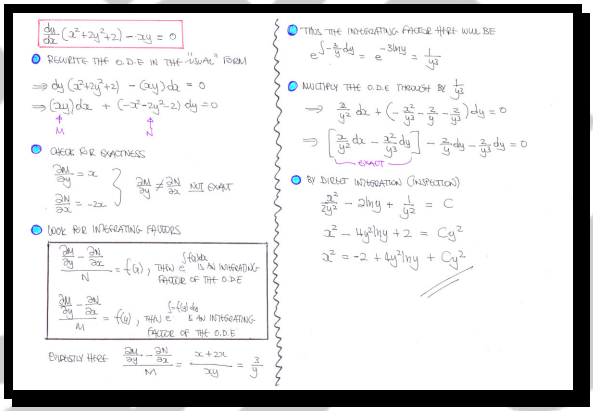


Question 10 (*+)**

Find a general solution of the following differential equation by looking for a suitable integrating factor.

$$\frac{dy}{dx}\left(x^2+2y^2+2\right)-xy=0\,.$$

$$\boxed{x^2=-2+4y^2\ln y+Cy^2}$$

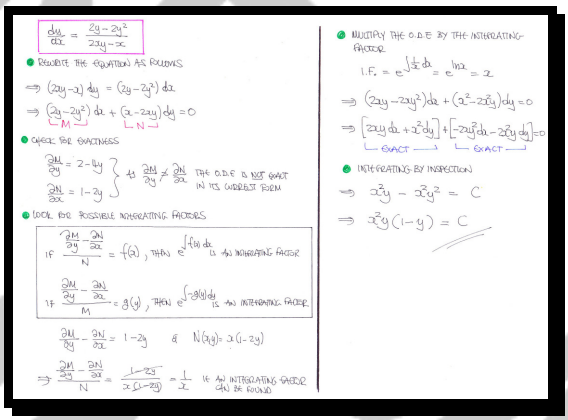


Question 11 (*+)**

Find a general solution of the following differential equation by looking for a suitable integrating factor.

$$\frac{dy}{dx}=\frac{2y-2y^2}{2xy-x}\,.$$

$$\boxed{x^2y(1-y)=C}$$

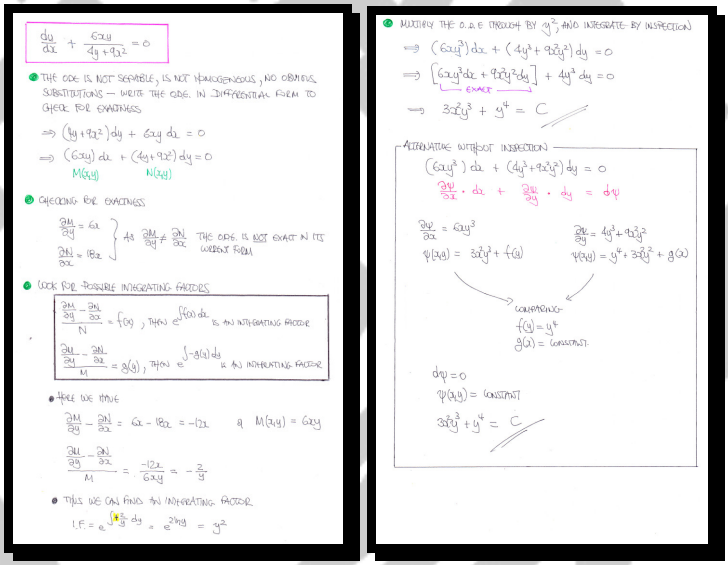


Question 12 (*+)**

Find a general solution of the following differential equation by looking for a suitable integrating factor.

$$\frac{dy}{dx}=\frac{6xy}{4y+9x^2}.$$

$$\boxed{3x^2y^3+y^4=C}$$

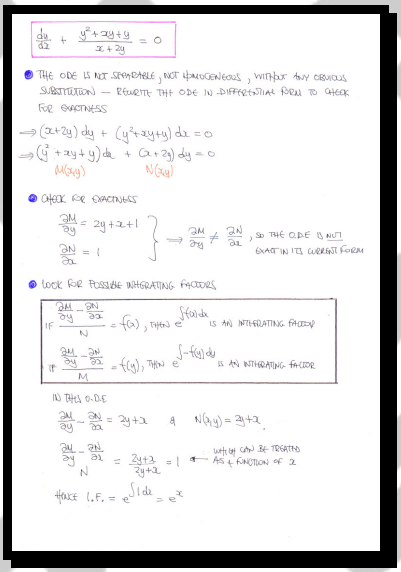


Question 13 (*+)**

Find a general solution of the following differential equation by looking for a suitable integrating factor.

$$\frac{dy}{dx}=\frac{y^2+xy+y}{x+2y}.$$

$$\boxed{y\mathrm{e}^x\left(y+x\right)=C}$$



$\frac{dy}{dx}+\frac{y^2+xy+y}{x+2y}=0$

- THE ODE IS NOT SEPARABLE, NOT HOMOGENEOUS, WITHOUT ANY OBVIOUS SUBSTITUTION — REWRITE THE ODE IN DIFFERENTIAL FORM TO CHECK FOR EXACTNESS

$\Rightarrow (x+2y)\,dy+(y^2+xy+y)\,dx=0$

$\Rightarrow (y^2+xy+y)\,dx+(x+2y)\,dy=0$

$M(x,y)$ $N(x,y)$

- CHECK FOR EXACTNESS

$\frac{\partial M}{\partial y}=2y+x+1$, $\frac{\partial N}{\partial x}=1$ $\Rightarrow \frac{\partial M}{\partial y}\neq\frac{\partial N}{\partial x}$, SO THE ODE IS NOT EXACT IN ITS CURRENT FORM

- LOOK FOR POSSIBLE INTEGRATING FACTORS

IF $\frac{\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}}{N}=f(x)$, THEN $\mathrm{e}^{\int f(x)\,dx}$ IS AN INTEGRATING FACTOR

IF $\frac{\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}}{M}=f(y)$, THEN $\mathrm{e}^{\int -f(y)\,dy}$ IS AN INTEGRATING FACTOR

IN THIS O.D.E

$\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}=2y+x$ & $N(x,y)=2y+x$.

$\frac{\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}}{N}=\frac{2y+x}{2y+x}=1$ ← WHICH CAN BE TREATED AS A FUNCTION OF x

HENCE $I.F.=\mathrm{e}^{\int 1\,dx}=\mathrm{e}^x$

- MULTIPLY THE O.D.E BY THE INTEGRATING FACTOR e^x

$\Rightarrow \mathrm{e}^x(y^2+xy+y)\,dx+\mathrm{e}^x(x+2y)\,dy=0$

EXACT (BY INSPECTION)

$\Rightarrow \mathrm{e}^x(y^2+xy)=C$

$\Rightarrow y\mathrm{e}^x(y+x)=C$

ALTERNATIVE WITHOUT INSPECTION

$\mathrm{e}^x(y^2+xy+y)\,dx+\mathrm{e}^x(x+2y)=0$

$(y\mathrm{e}^x+xy\mathrm{e}^x+y\mathrm{e}^x)\,dx+(x\mathrm{e}^x+2y\mathrm{e}^x)\,dy=0$

$\frac{\partial\phi}{\partial x}\cdot dx+\frac{\partial\phi}{\partial y}\cdot dy=d\phi$

THUS

- $\frac{\partial\phi}{\partial x}=y\mathrm{e}^x+xy\mathrm{e}^x+y\mathrm{e}^x$

BY PARTS IN x

$\phi(x,y)=y\mathrm{e}^x+y\left[x\mathrm{e}^x-\mathrm{e}^x\right]+y\mathrm{e}^x+f(y)$

$\phi(x,y)=y\mathrm{e}^x+xy\mathrm{e}^x-y\mathrm{e}^x+y\mathrm{e}^x+f(y)$

$\phi(x,y)=y\mathrm{e}^x+xy\mathrm{e}^x+f(y)$

- $\frac{\partial\phi}{\partial y}=x\mathrm{e}^x+y\mathrm{e}^x$

$\phi(x,y)=xy\mathrm{e}^x+y\mathrm{e}^x+g(x)$

COMPARING: $f(y)=g(x)=$ (CONSTANT)

- SO $d\phi=0$

$\phi(x,y)=$ CONSTANT

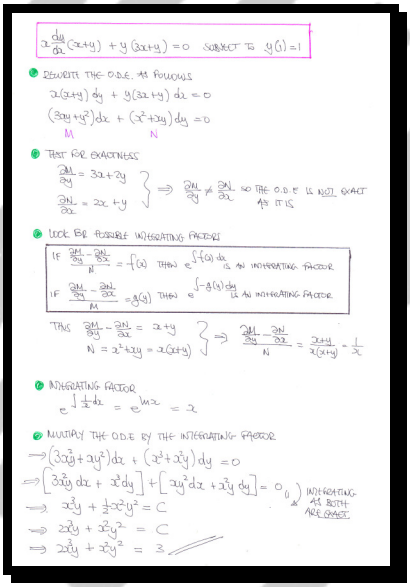
$\therefore xy\mathrm{e}^x+y\mathrm{e}^x=C$

Question 14 (*+)**

Determine the solution of the following differential equation by looking for a suitable integrating factor.

$$x\frac{dy}{dx}(x+y)+y(3x+y)=0, \quad y(1)=1.$$

$$\boxed{2yx^3+x^2y^2=3}$$

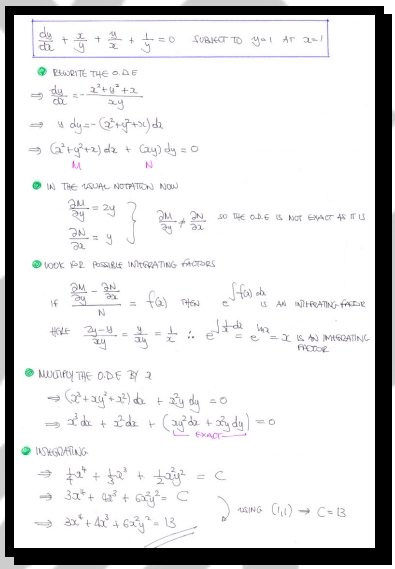


Question 15 (*+)**

Determine the solution of the following differential equation by looking for a suitable integrating factor.

$$\frac{dy}{dx}+\frac{x}{y}+\frac{y}{x}+\frac{1}{y}=0\,,\quad y(1)=1\,.$$

$$\boxed{3x^4+4x^3+6x^2y^2=13}$$

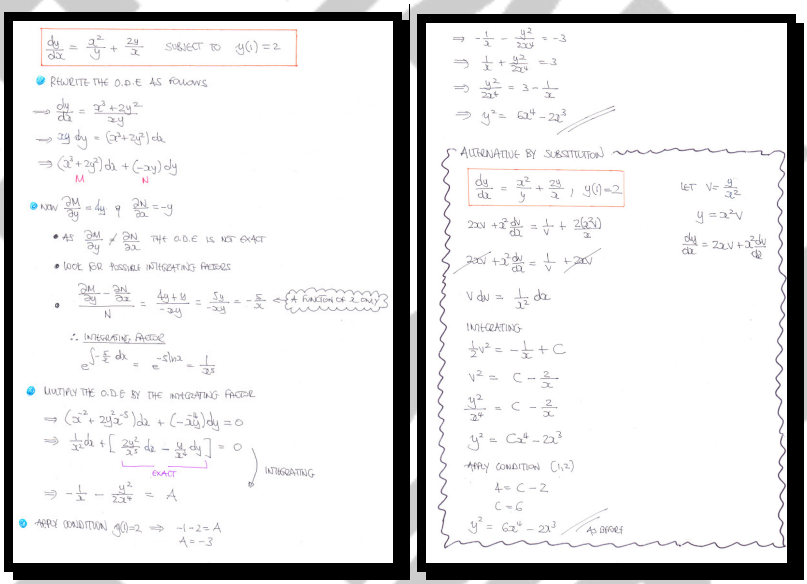


Question 16 (*+)**

Determine the solution of the following differential equation.

$$\frac{dy}{dx}=\frac{x^2}{y}+\frac{2y}{x}, \quad y(1)=2.$$

$$\boxed{y^2=6x^4-2x^3}$$

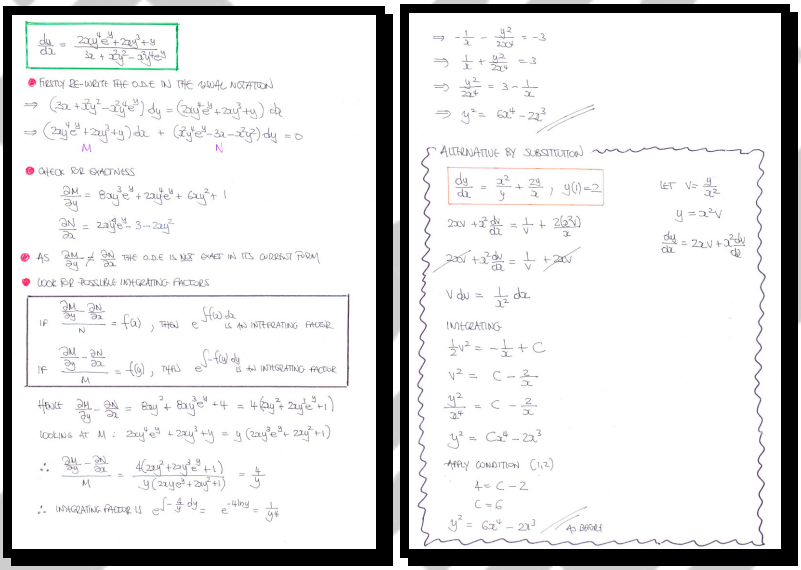


Question 17 (****)

Determine a general solution of the following differential equation by looking for a suitable integrating factor.

$$\frac{dy}{dx}=\frac{2xy^4\,\mathrm{e}^y+2xy^3+y}{3x+x^2y^2-x^2y^4\,\mathrm{e}^y}.$$

$$\boxed{x^2\,\mathrm{e}^y+\frac{x^2}{y}+\frac{x}{y^3}=C}$$



1ST ORDER BY VARIOUS TECHNIQUES

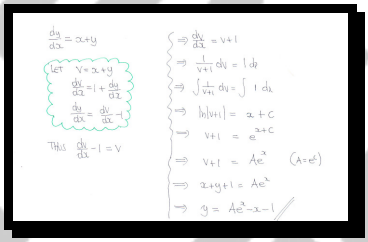
Question 1 (**)

By using a suitable substitution find a general solution of the differential equation

$$\frac{dy}{dx}=x+y\,,$$

giving the answer in the form $y=f\left(x\right)$.

$$\boxed{y=Ae^{x}-x-1}$$



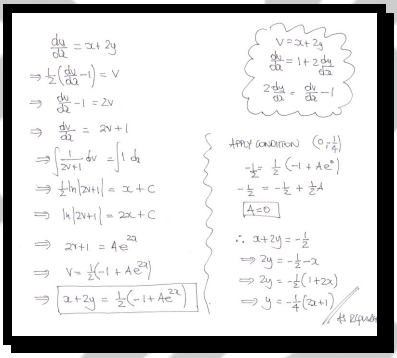
Question 2 (**)

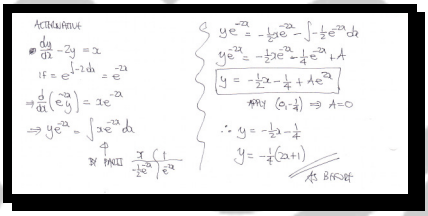
$$\frac{dy}{dx}=x+2y\,,\text{ with }\ y=-\frac{1}{4}\ \text{ at }\ x=0\,.$$

By using a suitable substitution, show that the solution of the differential equation is

$$y=-\frac{1}{4}\left(2x+1\right).$$

$$\boxed{\text{proof}}$$





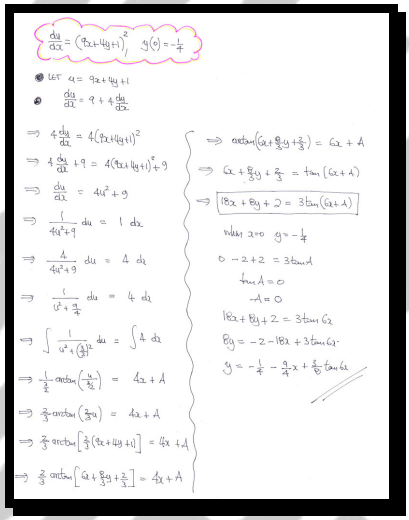
Question 3 (***)

Solve the differential equation

$$\frac{dy}{dx}=(9x+4y+1)^2,\quad y(0)=-\frac{1}{4}.$$

Give the answer in the form $y=f(x)$.

$$\boxed{y=-\frac{1}{4}-\frac{9}{4}x+\frac{3}{8}\tan 6x}$$

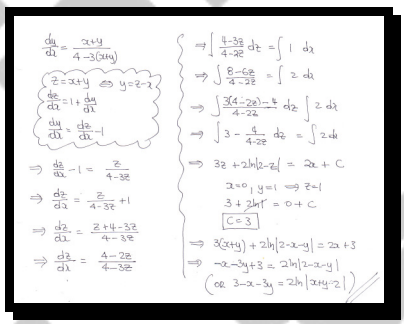


Question 4 (***)

Use a suitable substitution to solve the differential equation

$$\frac{dy}{dx}=\frac{x+y}{4-3(x+y)},\quad y(0)=1.$$

$$\boxed{2\ln|x+y-2|=3-x-3y}$$

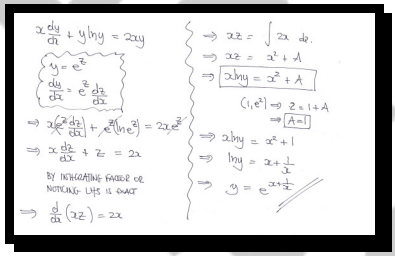


Question 5 (***)

Use the substitution $y = \mathrm{e}^z$ to solve the differential equation

$$x\frac{dy}{dx} + y\ln y = 2xy, \quad y(1) = \mathrm{e}^2.$$

$$\boxed{y = \mathrm{e}^{x+\frac{1}{x}}}$$



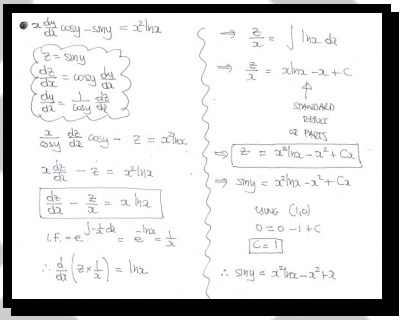
Question 6 (***)

Use the substitution $z = \sin y$ to solve the differential equation

$$x\frac{dy}{dx}\cos y - \sin y = x^2 \ln x, \quad y(1) = 0$$

subject to the condition $y = 0$ at $x = 1$.

$$\boxed{\sin y = x^2 \ln x - x^2 + x}$$

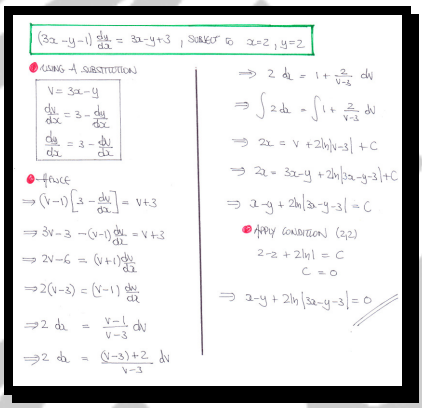


Question 7 (*+)**

Use a suitable substitution to find the solution of the following differential equation.

$$(3x-y-1)\frac{dy}{dx}=3x-y+3, \quad y(2)=2.$$

$$\boxed{x-y+2\ln|3x-y-3|=0}$$

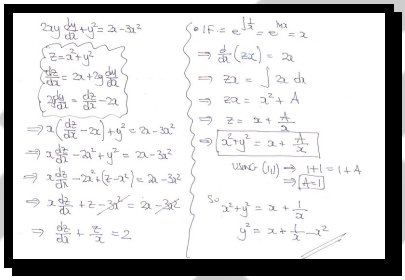


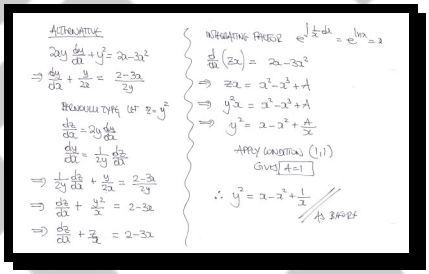
Question 8 (*+)**

By using the substitution $z=x^2+y^2$, or otherwise, solve the differential equation

$$2xy\frac{dy}{dx}+y^2=2x-3x^2, \quad y(1)=1.$$

$$\boxed{y^2=x-x^2+\frac{1}{x}}$$





Question 9 (*+)**

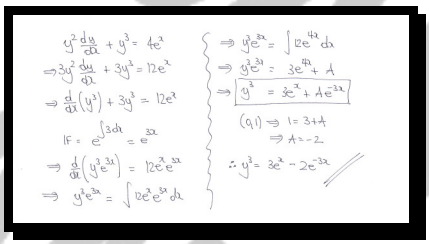
A curve with equation $y = f(x)$ passes through the point with coordinates $(0,1)$ and satisfies the differential equation

$$y^2 \frac{dy}{dx} + y^3 = 4\mathrm{e}^x .$$

By finding a suitable integrating factor, or otherwise, show that

$$y^3 = 3\mathrm{e}^x - 2\mathrm{e}^{-3x} .$$

proof



Question 10 (*+)**

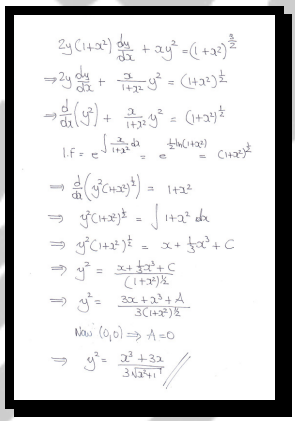
A curve with equation $y = f(x)$ passes through the origin and satisfies the differential equation

$$2y\left(1+x^2\right)\frac{dy}{dx}+xy^2=\left(1+x^2\right)^{\frac{3}{2}}.$$

By finding a suitable integrating factor, or otherwise, show that

$$y^2=\frac{x^3+3x}{3\sqrt{x^2+1}}.$$

proof



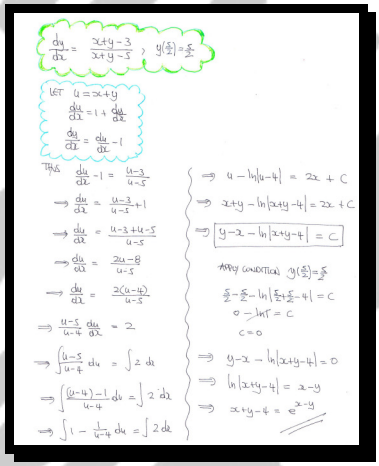
Question 11 (****)

Solve the differential equation

$$\frac{dy}{dx}=\frac{x+y-3}{x+y-5},$$

subject to the condition $y=\frac{5}{2}$ at $x=\frac{5}{2}$.

$$\boxed{x+y-4=\mathrm{e}^{x-y}}$$

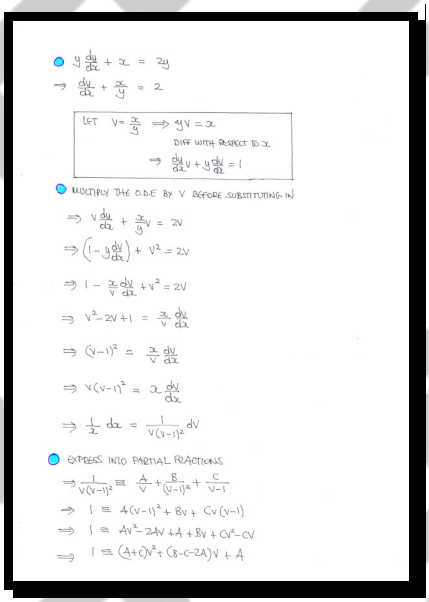


Question 12 (****)

Find a general solution of the following differential equation

$$y\frac{dy}{dx}+x=2y\,.$$

$$\boxed{(x-y)\mathrm{e}^{\frac{y}{x-y}}=C}$$



$y\frac{dy}{dx}+x=2y$

$\Rightarrow \frac{dy}{dx}+\frac{x}{y}=2$

LET $V=\frac{x}{y} \Rightarrow yV=x$

DIFF WITH RESPECT TO x

$\Rightarrow \frac{dy}{dx}V+y\frac{dV}{dx}=1$

MULTIPLY THE O.D.E BY V BEFORE SUBSTITUTING IN

$\Rightarrow V\frac{dy}{dx}+\frac{x}{y}V=2V$

$\Rightarrow \left(1-y\frac{dV}{dx}\right)+V^2=2V$

$\Rightarrow 1-\frac{x}{V}\frac{dV}{dx}+V^2=2V$

$\Rightarrow V^2-2V+1=\frac{x}{V}\frac{dV}{dx}$

$\Rightarrow (V-1)^2=\frac{x}{V}\frac{dV}{dx}$

$\Rightarrow V(V-1)^2=x\frac{dV}{dx}$

$\Rightarrow \frac{1}{x}dx=\frac{1}{V(V-1)^2}dV$

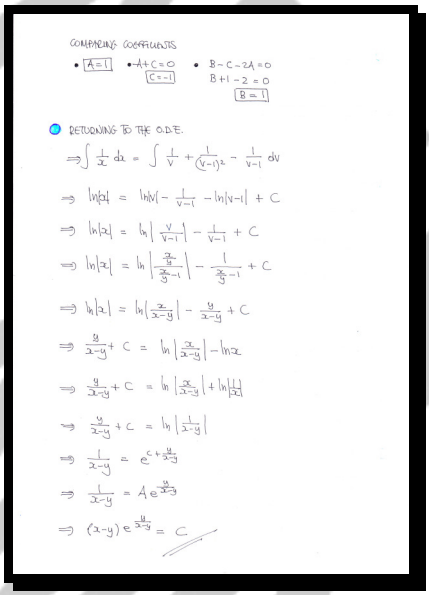
EXPRESS INTO PARTIAL FRACTIONS

$\Rightarrow \frac{1}{V(V-1)^2}\equiv \frac{A}{V}+\frac{B}{(V-1)^2}+\frac{C}{V-1}$

$\Rightarrow 1\equiv A(V-1)^2+BV+CV(V-1)$

$\Rightarrow 1\equiv AV^2-2AV+A+BV+CV^2-CV$

$\Rightarrow 1\equiv (A+C)V^2+(B-C-2A)V+A$



COMPARING COEFFICIENTS

• $A=1$ • $A+C=0$, $C=-1$ • $B-C-2A=0$, $B+1-2=0$, $B=1$

RETURNING TO THE O.D.E.

$\Rightarrow \int \frac{1}{x}dx=\int \frac{1}{V}+\frac{1}{(V-1)^2}-\frac{1}{V-1}dV$

$\Rightarrow \ln|x|=\ln|V|-\frac{1}{V-1}-\ln|V-1|+C$

$\Rightarrow \ln|x|=\ln\left|\frac{V}{V-1}\right|-\frac{1}{V-1}+C$

$\Rightarrow \ln|x|=\ln\left|\frac{\frac{x}{y}}{\frac{x}{y}-1}\right|-\frac{1}{\frac{x}{y}-1}+C$

$\Rightarrow \ln|x|=\ln\left|\frac{x}{x-y}\right|-\frac{y}{x-y}+C$

$\Rightarrow \frac{y}{x-y}+C=\ln\left|\frac{x}{x-y}\right|-\ln x$

$\Rightarrow \frac{y}{x-y}+C=\ln\left|\frac{x}{x-y}\right|+\ln\left|\frac{1}{x}\right|$

$\Rightarrow \frac{y}{x-y}+C=\ln\left|\frac{1}{x-y}\right|$

$\Rightarrow \frac{1}{x-y}=e^{C+\frac{y}{x-y}}$

$\Rightarrow \frac{1}{x-y}=Ae^{\frac{y}{x-y}}$

$\Rightarrow (x-y)e^{\frac{y}{x-y}}=C$

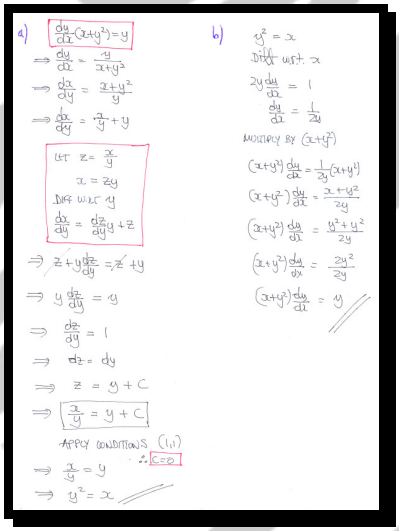
Question 13 (****)

$$\frac{dy}{dx}\left(x+y^2\right)=y\,.$$

a) Solve the above differential equation, subject to $y=1$ at $x=1$ by considering $\dfrac{dx}{dy}$, followed by a suitable substitution..

b) Verify the validity of the answer obtained in part **(a)**.

$$\boxed{y^2=x}$$



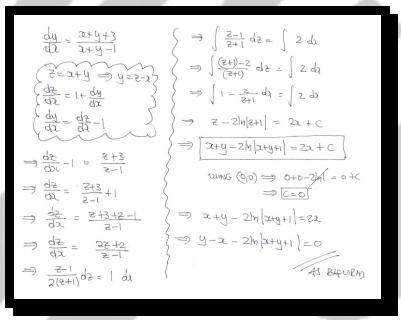
Question 14 (****)

$$\frac{dy}{dx}=\frac{x+y+3}{x+y-1},\ y(0)=0.$$

Show that the solution of the above differential equation is

$$y-x-2\ln(x+y+1)=0.$$

proof



Y.G.B. FREE SAMPLE T. MADAS

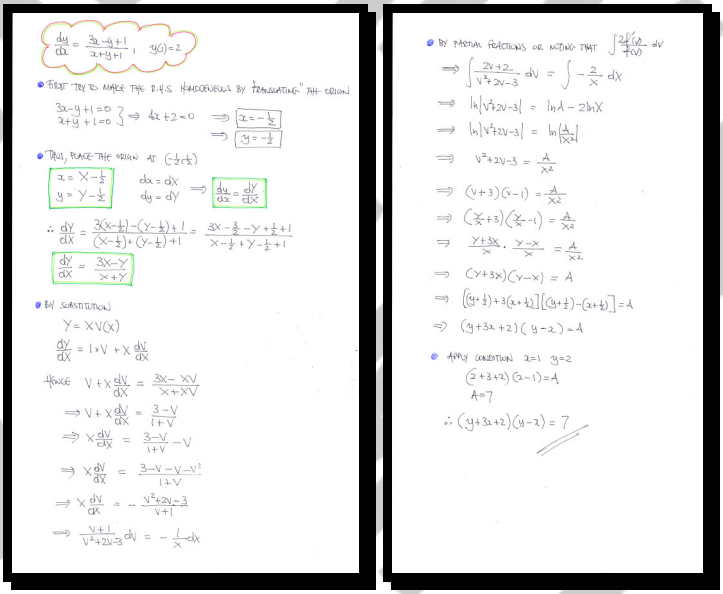
Question 15 (**+)**

$$\frac{dy}{dx}=\frac{3x-y+1}{x+y+1},\ y(1)=2.$$

Solve the differential equation to show that

$$(y-x)(y+3x+2)=7.$$

proof



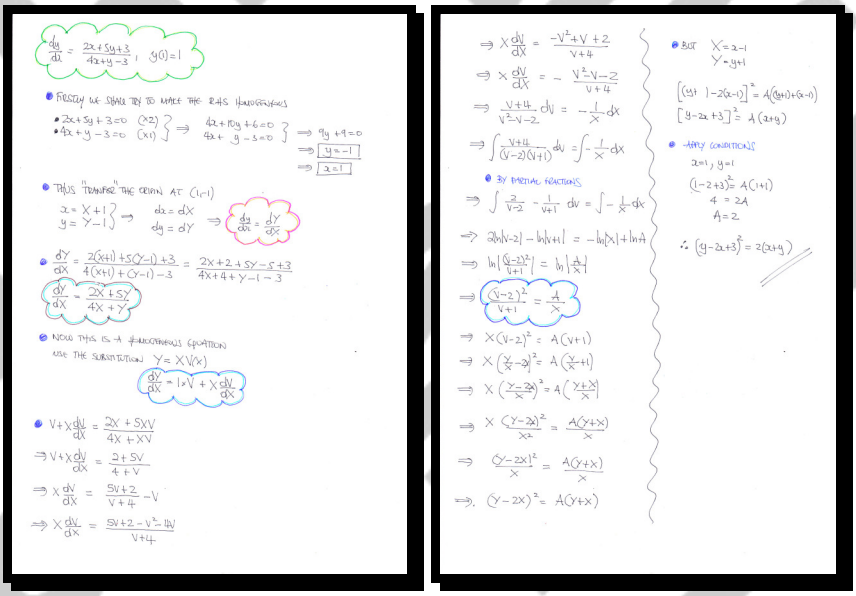
Question 16 (**+)**

$$\frac{dy}{dx}=\frac{2x+5y+3}{4x+y-3},\ y(1)=1.$$

Solve the differential equation to show that

$$(y-2x+3)^2=2(x+y).$$

proof



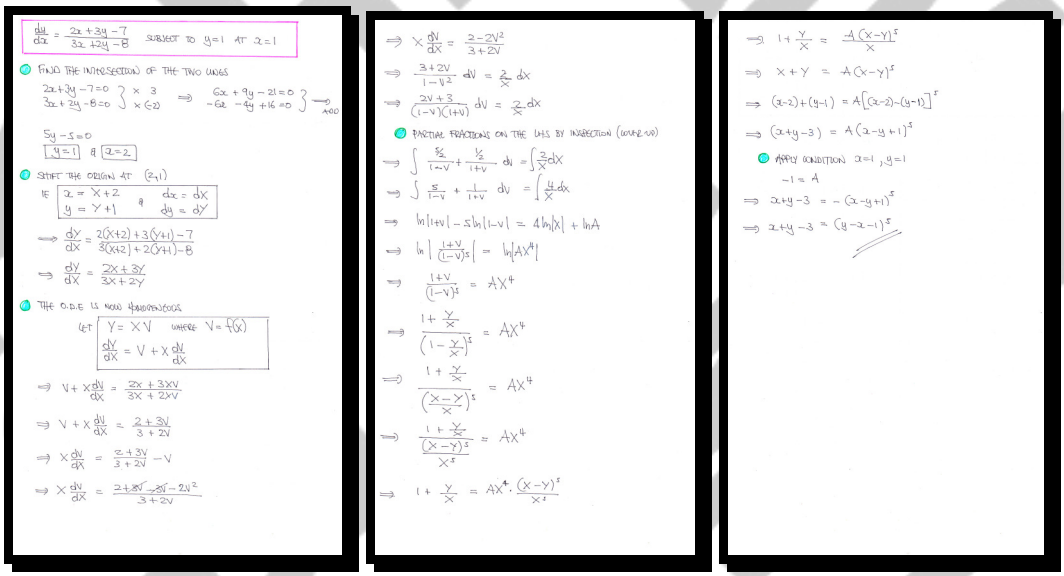
Question 17 (**+)**

Solve the following differential equation

$$\frac{dy}{dx}=\frac{2x+3y-7}{3x+2y-8},\quad y(1)=1$$

Give the answer in the form $(y-x-1)^5=f(x,y)$, where $f(x,y)$ is a function to be found.

$$(y-x-1)^5=y+x-3$$



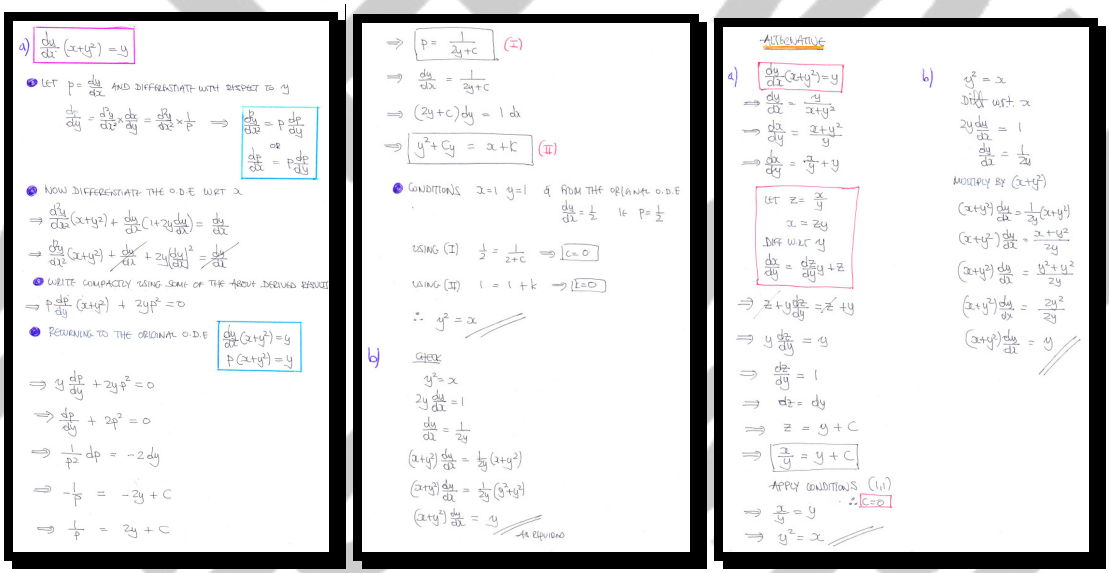
Question 18 (**+)**

$$\frac{dy}{dx}\left(x+y^2\right)=y\,.$$

a) Solve the above differential equation, subject to $y=1$ at $x=1$.

b) Verify the validity of the answer obtained in part **(a)**.

$$\boxed{y^2=x}$$



Question 19 (**+)**

Find a general solution for the following differential equation

$$\frac{dy}{dx}=\frac{1+y^2}{-x+\arctan y}.$$

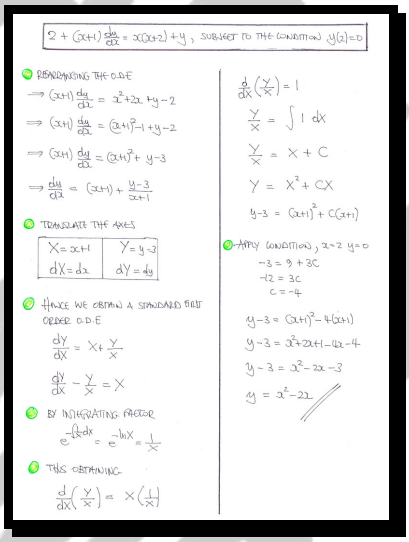
$$\boxed{x=-1+\arctan y+A\mathrm{e}^{-\arctan y}}$$

Question 20 (**+)**

$$2+(x+1)\frac{dy}{dx}=x(x+2)+y\,.$$

Solve the above differential equation, subject to $y(2)=0$.

$$\boxed{y=x^2-2x}$$



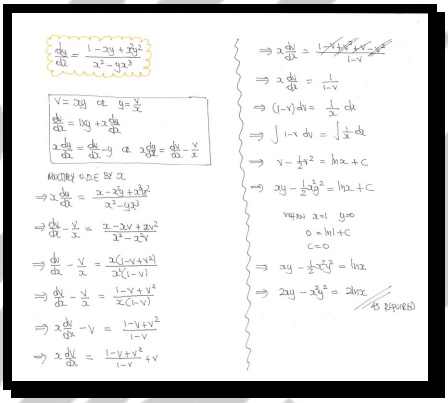
Question 21 (*****)

Solve the differential equation

$$\frac{dy}{dx}=\frac{1-xy+x^2y^2}{x^2-yx^3},\ x>0,$$

subject to the condition $y(1)=0$.

$$\boxed{2xy-x^2y^2=2\ln x}$$

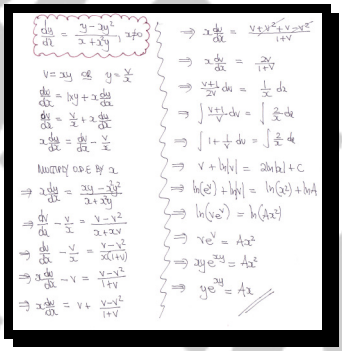


Question 22 (*****)

Find a general solution for the differential equation

$$\frac{dy}{dx}=\frac{y-xy^2}{x+yx^2}\,,\ x\neq 0\,.$$

$$\boxed{y\,\mathrm{e}^{xy}=Cx}$$



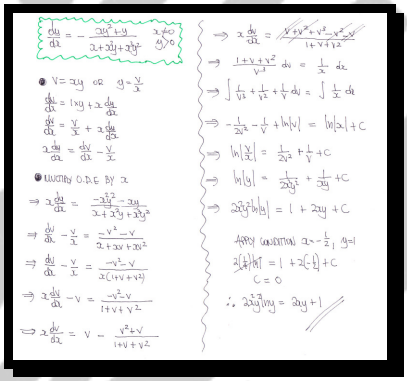
Question 23 (*****)

Solve the differential equation

$$\frac{dy}{dx}=-\frac{xy^2+y}{x+yx^2+x^3y^2},\ x\neq 0,\ y>0,$$

subject to the condition $y\left(\frac{1}{2}\right)=1$.

$$\boxed{2x^2y^2\ln y=2xy+1}$$



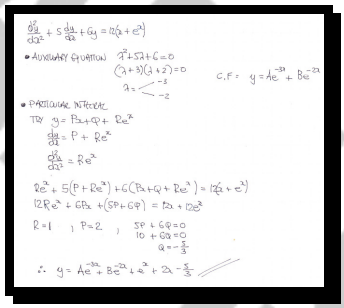
2^{ND} ORDER WITH CONSTANT COEFFICIENTS

Question 1 (**)

Find a general solution of the differential equation

$$\frac{d^2y}{dx^2}+5\frac{dy}{dx}+6y=12\left(x+\mathrm{e}^x\right).$$

$$\boxed{y=A\mathrm{e}^{-3x}+B\mathrm{e}^{-2x}+\mathrm{e}^x+2x-\frac{5}{3}}$$

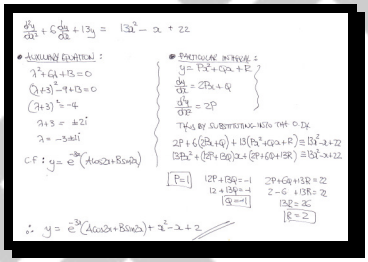


Question 2 (**)

Find a general solution of the differential equation

$$\frac{d^2y}{dx^2}+6\frac{dy}{dx}+13y=13x^2-x+22.$$

$$\boxed{y=\mathrm{e}^{-3x}\left(A\cos 2x+B\sin 2x\right)+x^2-x+2}$$



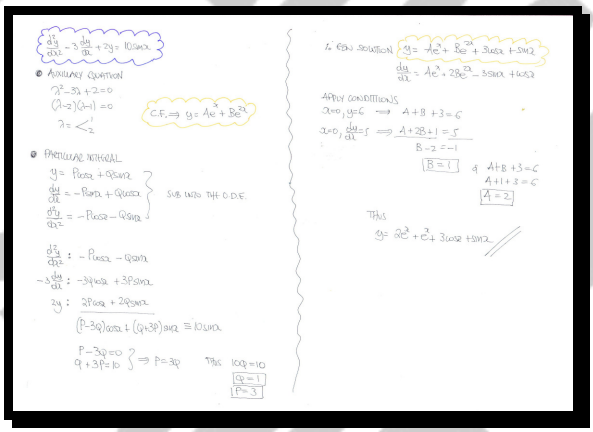
Question 3 (**)

Find a solution of the differential equation

$$\frac{d^2y}{dx^2}-3\frac{dy}{dx}+2y=10\sin x,$$

subject to the boundary conditions $y=6$ and $\frac{dy}{dx}=5$ at $x=0$.

$$\boxed{y=2\mathrm{e}^{x}+\mathrm{e}^{2x}+3\cos x+\sin x}$$

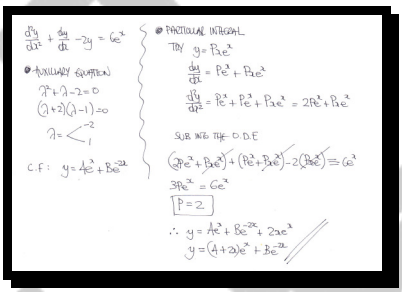


Question 4 (**)

Find a general solution of the differential equation

$$\frac{d^2y}{dx^2}+\frac{dy}{dx}-2y=6\mathrm{e}^{x}.$$

$$\boxed{y=(A+2x)\mathrm{e}^{x}+B\mathrm{e}^{-2x}}$$



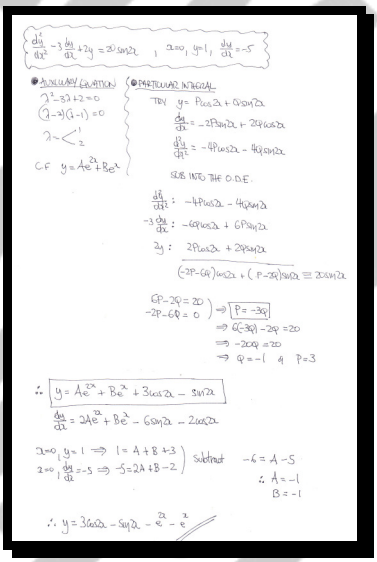
Question 5 (**)

Find a solution of the differential equation

$$\frac{d^2y}{dx^2}-3\frac{dy}{dx}+2y=20\sin 2x\,,$$

subject to the boundary conditions $y=1$ and $\frac{dy}{dx}=-5$ at $x=0$.

$$\boxed{y=3\cos 2x-\sin 2x-\mathrm{e}^{2x}-\mathrm{e}^{x}}$$

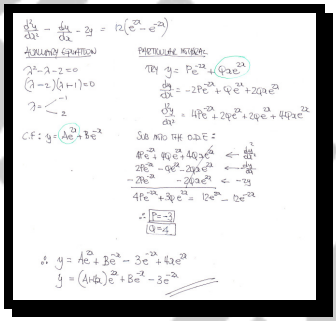


Question 6 (**)

Find a general solution of the differential equation

$$\frac{d^2y}{dx^2}-\frac{dy}{dx}-2y=12\left(\mathrm{e}^{2x}-\mathrm{e}^{-2x}\right).$$

$$\boxed{y=\left(A+4x\right)\mathrm{e}^{2x}+B\mathrm{e}^{-x}-3\mathrm{e}^{-2x}}$$



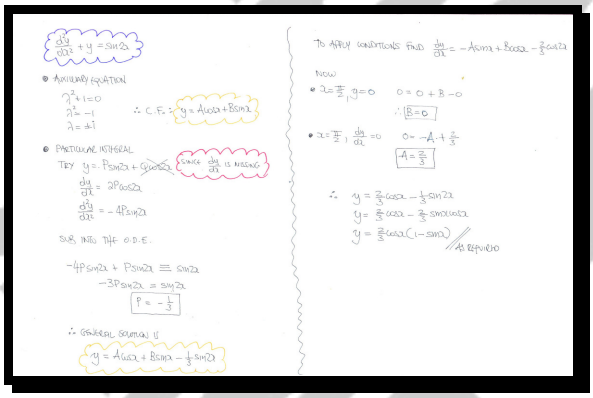
Question 7 (**)

$$\frac{d^2y}{dx^2}+y=\sin 2x, \quad \text{with } y=0 \text{ , } \frac{dy}{dx}=0 \text{ at } x=\frac{\pi}{2}.$$

Show that a solution of the above differential equation is

$$y=\frac{2}{3}\cos x\left(1-\sin x\right).$$

proof



Question 8 (+)**

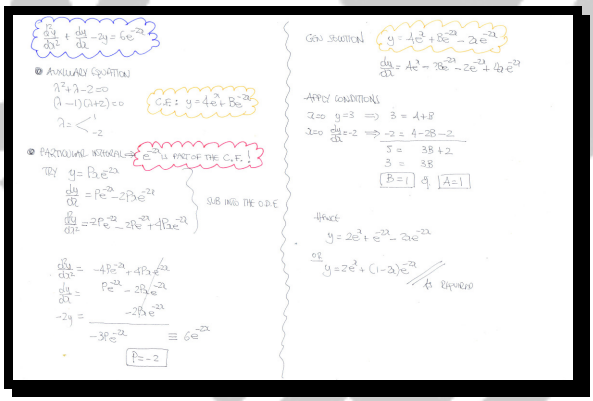
$$\frac{d^2y}{dx^2}+\frac{dy}{dx}-2y=6\mathrm{e}^{-2x},$$

with $y=3$ and $\frac{dy}{dx}=-2$ at $x=0$.

Show that the solution of the above differential equation is

$$y=2\mathrm{e}^{x}+(1-2x)\mathrm{e}^{-2x}.$$

proof

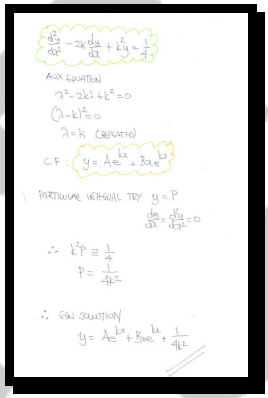


Question 9 (+)**

Find a general solution of the differential equation

$$\frac{d^2y}{dx^2}-2k\frac{dy}{dx}+k^2y=\frac{1}{4},\ k>0.$$

$$\boxed{y=A\mathrm{e}^{kx}+Bx\mathrm{e}^{kx}+\frac{1}{4k^2}}$$



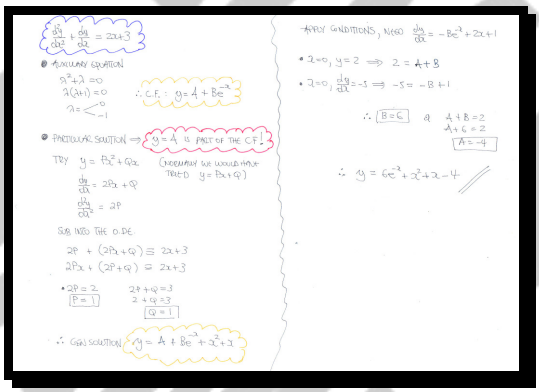
Question 10 (+)**

Find the solution of the differential equation

$$\frac{d^2y}{dx^2}+\frac{dy}{dx}=2x+3,$$

subject to the conditions $y=2$, $\frac{dy}{dx}=-5$ at $x=0$.

$$\boxed{y=x^2+x-4+6\mathrm{e}^{-x}}$$



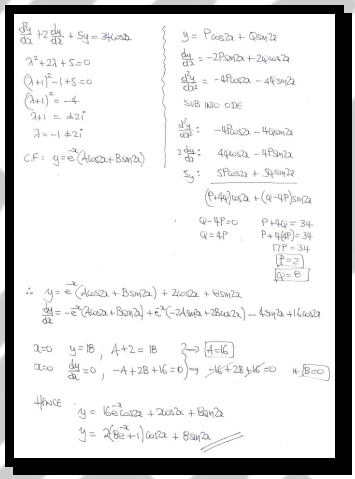
Question 11 (+)**

Find a solution of the differential equation

$$\frac{d^2y}{dx^2}+2\frac{dy}{dx}+5y=34\cos 2x,$$

subject to the boundary conditions $y=18$ and $\frac{dy}{dx}=0$ at $x=0$.

$$\boxed{y=2\left(8\mathrm{e}^{-x}+1\right)\cos 2x+8\sin 2x}$$



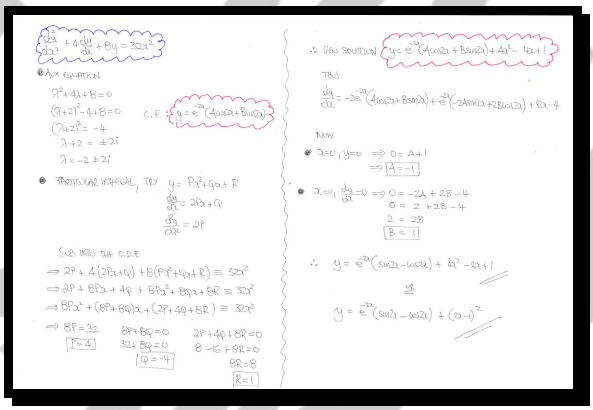
Question 12 (+)**

The curve C has a local minimum at the origin and satisfies the differential equation

$$\frac{d^2y}{dx^2}+4\frac{dy}{dx}+8y=32x^2 .$$

Find an equation for C.

$$\boxed{y=\mathrm{e}^x\left(\sin 2x+\cos 2x\right)+\left(2x-1\right)^2}$$



Question 13 (+)**

$$\frac{d^2x}{dt^2}+9x+12\sin 3t=0\,,\ t\geq 0\,,$$

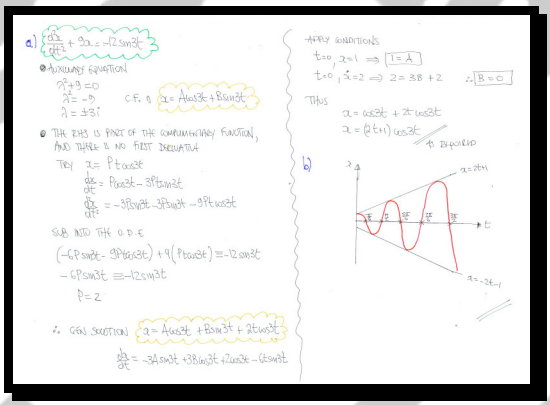
with $x=1$, $\dfrac{dx}{dt}=2$ at $t=0$.

a) Show that a solution of the differential equation is

$$x=(2t+1)\cos 3t\,.$$

b) Sketch the graph of x.

proof



Question 14 (+)**

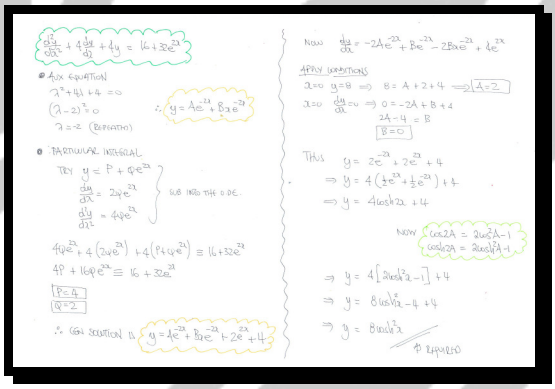
$$\frac{d^2y}{dx^2}+4\frac{dy}{dx}+4y=16+32\mathrm{e}^{2x},$$

with $y=8$ and $\frac{dy}{dx}=0$ at $x=0$.

Show that the solution of the above differential equation is

$$y=8\cosh^2 x.$$

proof



Question 15 (+)**

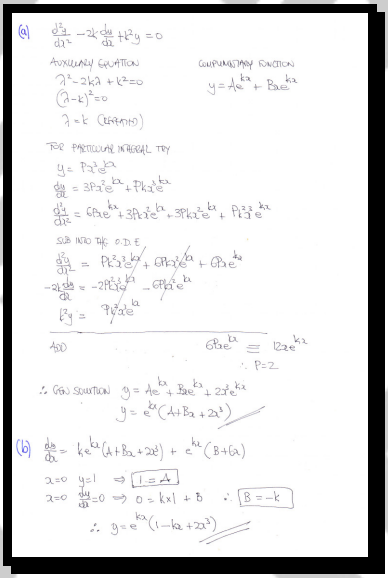
$$\frac{d^2y}{dx^2}-2k\frac{dy}{dx}+k^2y=12x\,\mathrm{e}^{kx},\ k>0$$

a) Find a general solution of the differential equation given that $y=Px^3\,\mathrm{e}^{kx}$, where P is a constant, is part of the solution.

b) Given further that $y=1$, $\frac{dy}{dx}=0$ at $x=0$ show that

$$y=\mathrm{e}^{kx}\left(2x^3-kx+1\right).$$

$$\boxed{y=\mathrm{e}^{kx}\left(2x^3+Ax+B\right)}$$



(a) $\frac{d^2y}{dx^2}-2k\frac{dy}{dx}+k^2y=0$

AUXILIARY EQUATION $\lambda^2-2k\lambda+k^2=0$, $(\lambda-k)^2=0$, $\lambda=k$ (REPEATED)

COMPLEMENTARY FUNCTION $y=Ae^{kx}+Bxe^{kx}$

TRY PARTICULAR INTEGRAL $y=Px^3e^{kx}$

$\frac{dy}{dx}=3Px^2e^{kx}+Pkx^3e^{kx}$

$\frac{d^2y}{dx^2}=6Pxe^{kx}+3Pkx^2e^{kx}+3Pkx^2e^{kx}+Pk^2x^3e^{kx}$

SUB INTO THE O.D.E

$\frac{d^2y}{dx^2}=Pk^2x^3e^{kx}+6Pkx^2e^{kx}+6Pxe^{kx}$

$-2k\frac{dy}{dx}=-2Pkx^3e^{kx}-6Pkx^2e^{kx}$

$k^2y=Pk^2x^3e^{kx}$

ADD $\quad 6Pxe^{kx}\equiv 12xe^{kx}$ $\therefore P=2$

$\therefore$ GEN SOLUTION $y=Ae^{kx}+Bxe^{kx}+2x^3e^{kx}$

$y=e^{kx}(A+Bx+2x^3)$

(b) $\frac{dy}{dx}=ke^{kx}(A+Bx+2x^3)+e^{kx}(B+6x)$

$x=0$ $y=1$ $\Rightarrow$ $\boxed{1=A}$

$x=0$ $\frac{dy}{dx}=0$ $\Rightarrow$ $0=kx1+B$ $\therefore$ $\boxed{B=-k}$

$\therefore\ y=e^{kx}(1-kx+2x^3)$

Question 16 (+)**

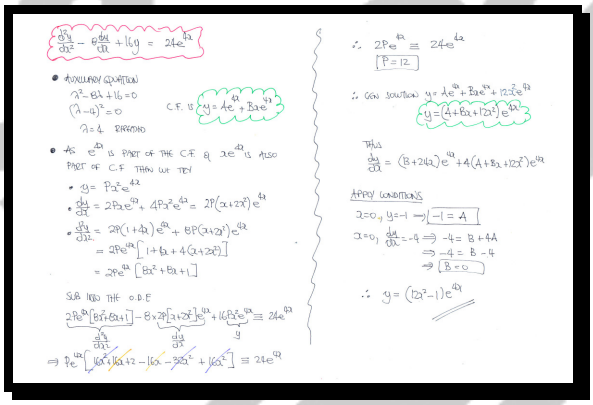
Show that the solution of the differential equation

$$\frac{d^2y}{dx^2}-8\frac{dy}{dx}+16y=24\mathrm{e}^{4x},$$

subject to the boundary conditions $y=-1$, $\frac{dy}{dx}=-4$ at $x=0$, can be written as

$$y=\left(12x^2-1\right)\mathrm{e}^{4x}.$$

proof



Question 17 (***)

$$\frac{d^2y}{dx^2}-6\frac{dy}{dx}+9y=4\mathrm{e}^{3x}.$$

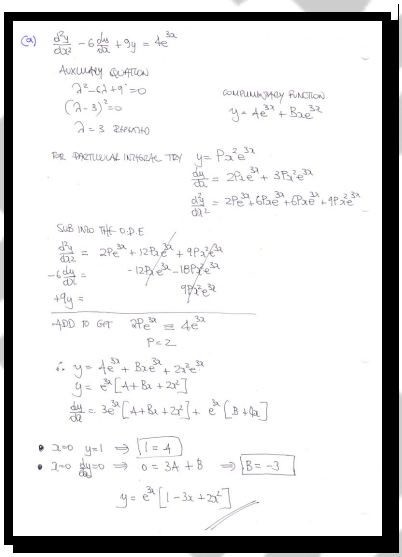
a) Find a solution of the differential equation given that $y=1$, $\dfrac{dy}{dx}=0$ at $x=0$.

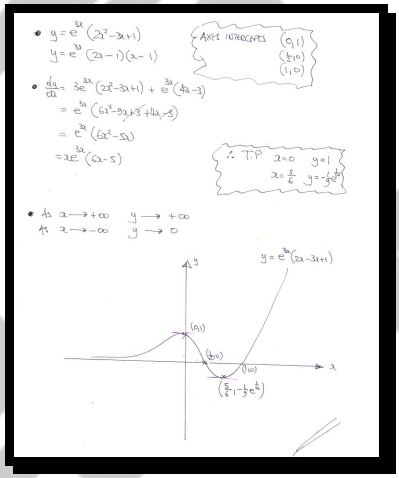
b) Sketch the graph of y.

The sketch must include …

- the coordinates of any points where the graph meets the coordinate axes.
- the coordinates of any stationary points of the curve.
- clear indications of how the graph looks for large positive or negative values of x.

$$\boxed{y=\mathrm{e}^{3x}\left(2x^2-3x+1\right)}$$





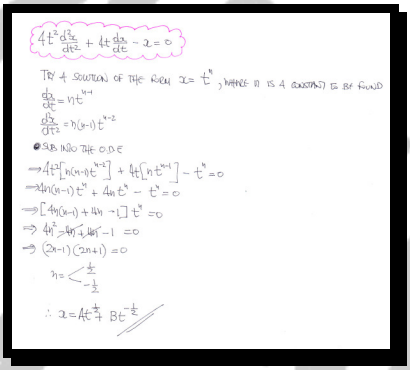
2^{ND} ORDER or HIGHER CAUCHY EULER TYPE

Question 1 (**)

Find the general solution of the following differential equation.

$$4t^2\frac{d^2x}{dt^2}+4t\frac{dx}{dt}-x=0\,.$$

$$\boxed{x=At^{\frac{1}{2}}+Bt^{-\frac{1}{2}}}$$

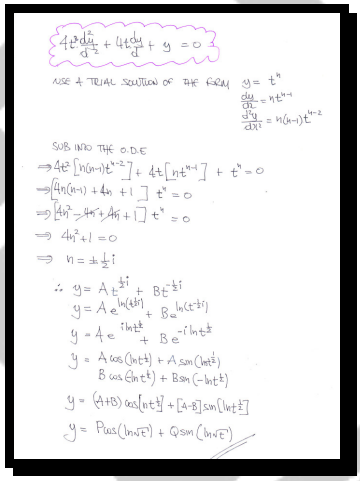


Question 2 (+)**

Find the general solution of the following differential equation.

$$4t^2\frac{d^2y}{dt^2}+4t\frac{dy}{dt}+y=0\,.$$

$$\boxed{y=P\cos\left[\ln\sqrt{t}\right]+P\sin\left[\ln\sqrt{t}\right]}$$



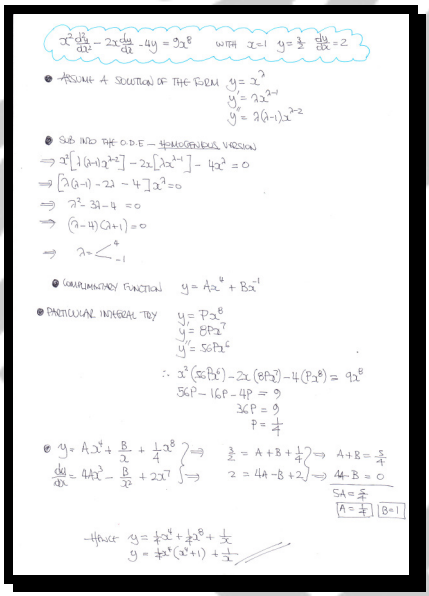
Question 3 (***)

$$x^2\frac{d^2y}{dx^2}-2x\frac{dy}{dx}-4y=9x^8.$$

Determine the solution of the above differential equation subject to the boundary conditions

$$y=\frac{3}{2},\ \frac{dy}{dx}=2 \text{ at } x=1.$$

$$\boxed{y=\frac{1}{4}x^4\left(x^4+1\right)+\frac{1}{x}}$$

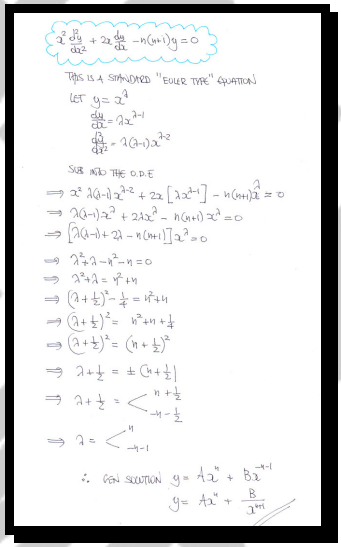


Question 4 (***)

Find the general solution of the following differential equation

$$x^2\frac{d^2y}{dx^2}+2x\frac{dy}{dx}-n(n+1)y=0\,.$$

$$\boxed{y=Ax^n+\frac{B}{x^{n+1}}}$$



Question 5 (*+)**

Given that if $x = \mathrm{e}^t$ and $y = f(x)$, show clearly that …

a) … $x\dfrac{dy}{dx} = \dfrac{dy}{dt}$.

b) … $x^2\dfrac{d^2y}{dx^2} = \dfrac{d^2y}{dt^2} - \dfrac{dy}{dt}$.

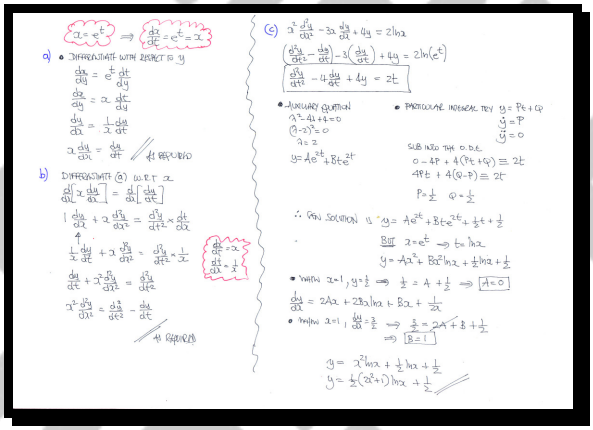
The following differential equation is to be solved

$$x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 4y = 2\ln x$$

subject to the boundary conditions $y = \dfrac{1}{2}, \dfrac{dy}{dx} = \dfrac{3}{2}$ at $x = 1$.

c) Use the substitution $x = \mathrm{e}^t$ to solve the above differential equation.

$$\boxed{y = \frac{1}{2} + \frac{1}{2}\left(2x^2 + 1\right)\ln x}$$

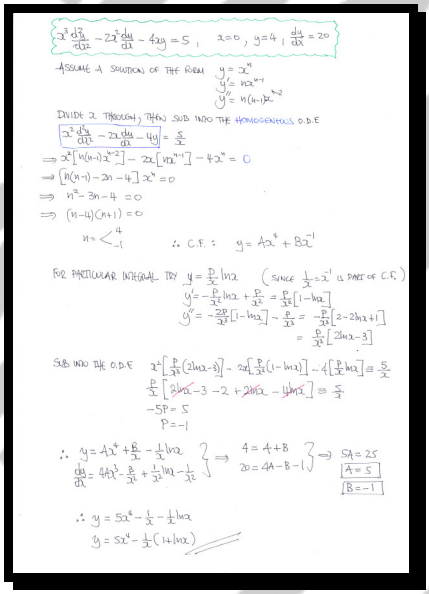


Question 6 (*+)**

$$x^3\frac{d^2y}{dx^2}-2x^2\frac{dy}{dx}-4xy=5.$$

Find the solution of the above differential equation subject to the boundary conditions $y=4$, $\dfrac{dy}{dx}=20$ at $x=0$.

$$\boxed{y=5x^4-\frac{1}{x}(1+\ln x)}$$

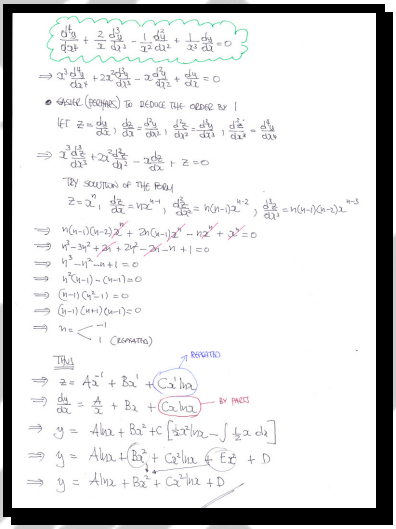


Question 7 (****)

Find the general solution of the following differential equation

$$\frac{d^4y}{dx^4}+\frac{2}{x}\frac{d^3y}{dx^3}-\frac{1}{x^2}\frac{d^2y}{dx^2}+\frac{1}{x^3}\frac{dy}{dx}=0\,.$$

$$\boxed{y=A\ln x+Bx^2+Cx^2\ln x+D}$$



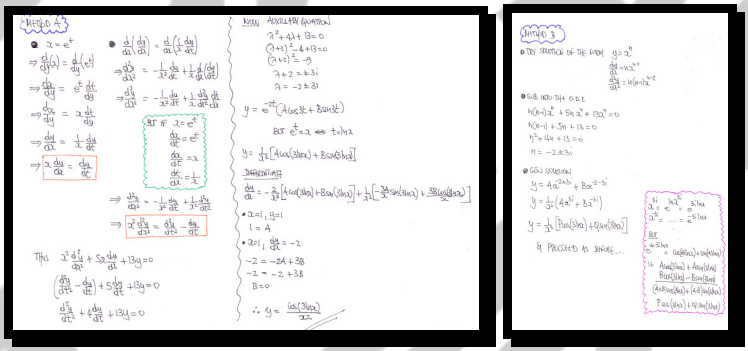
Question 8 (****)

The curve with equation $y = f(x)$ satisfies

$$x^2 \frac{d^2y}{dx^2} + 5x\frac{dy}{dx} + 13y = 0, \ x > 0.$$

By using the substitution $x = e^t$, or otherwise, determine an equation for $y = f(x)$, given further that $y = 1$ and $\frac{dy}{dx} = -2$ at $x = 1$.

$$\boxed{y = \frac{\cos(3\ln x)}{x^2}}$$



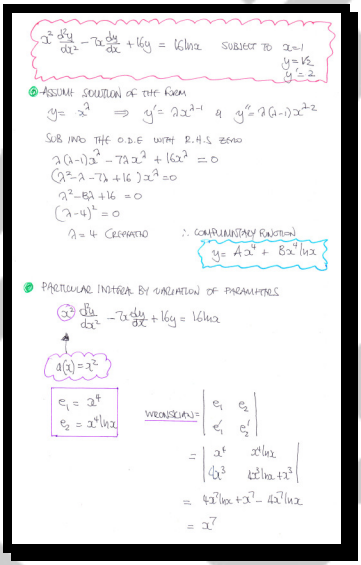
Question 9 (****)

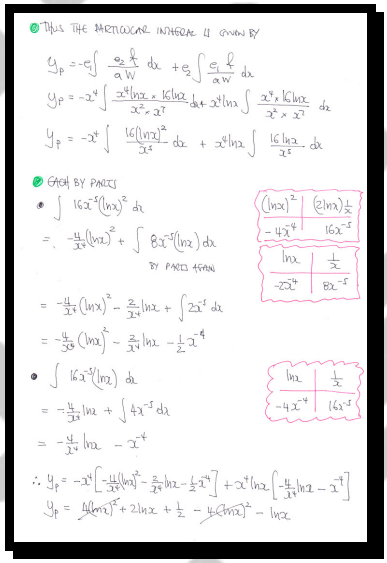
Use variation of parameters to determine the specific solution of the following differential equation

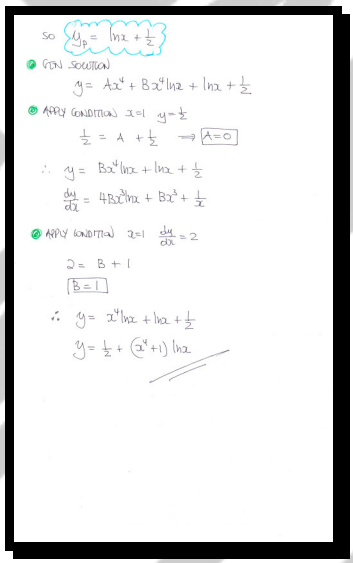
$$x^2\frac{d^2y}{dx^2}-7x\frac{dy}{dx}+16y=16\ln x\,,$$

given further that $y=\frac{1}{2}$, $\frac{dy}{dx}=2$ at $x=1$.

$$\boxed{y=\frac{1}{2}+\left(1+x^4\right)\ln x}$$







2^{ND} ORDER ODEs WITH MISSING INDEPENDENT VARIABLE

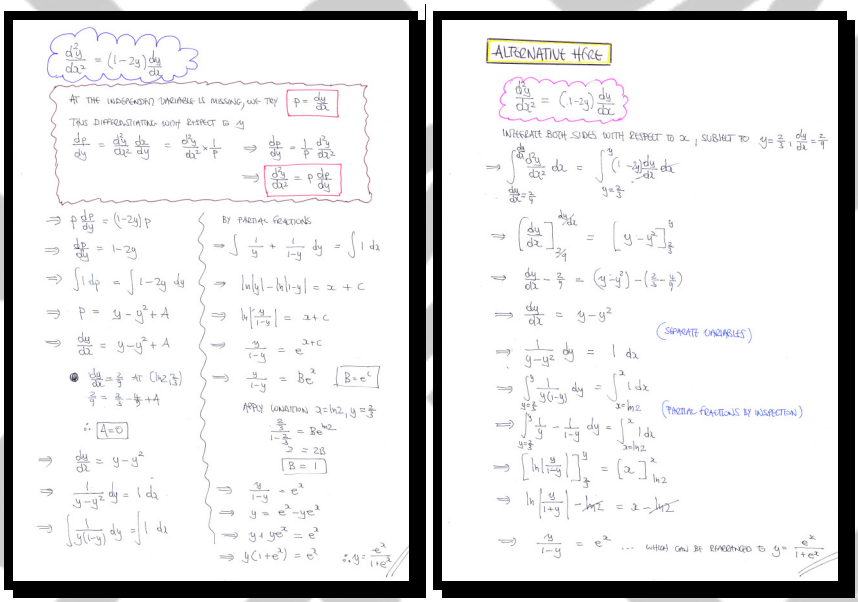
Question 1 **(****+)**

The curve C, has gradient $\frac{2}{9}$ at the point with coordinates $\left(\ln 2, \frac{2}{3}\right)$, and satisfies the differential relationship

$$\frac{d^2y}{dx^2} = (1-2y)\frac{dy}{dx}, \qquad y < \frac{1}{2}.$$

Find an equation for C, giving the answer in the form $y = f(x)$.

$$\boxed{y = \frac{e^x}{1+e^x} = \frac{1}{e^x + e^{-x}} = \frac{1}{2}\operatorname{sech} x}$$



Question 2 (**+)**

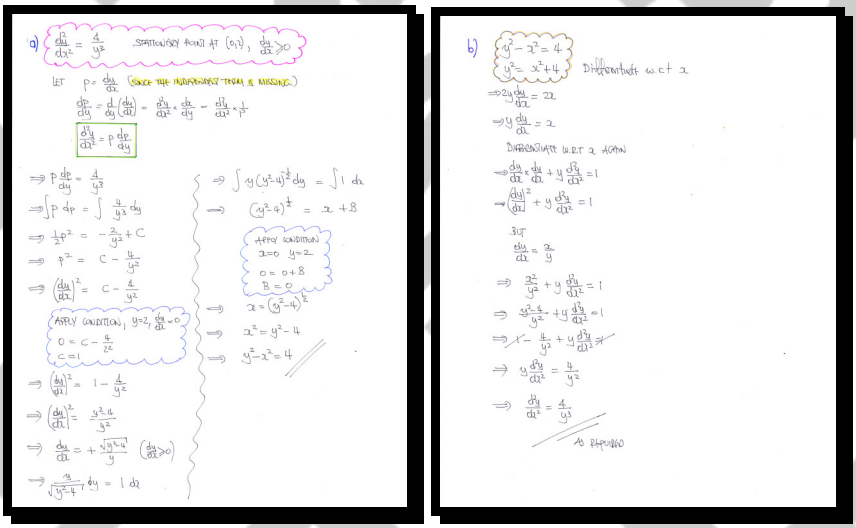
The curve C, has a stationary point at $(0,2)$ and satisfies the differential relationship

$$\frac{d^2y}{dx^2}=\frac{4}{y^3},\ y\neq 0.$$

a) Given further that $\frac{dy}{dx}\geq 0$ along C, determine a simplified expression for the Cartesian equation of C.

b) Verify by differentiation the answer to part **(a)**.

$$y^2-x^2=4$$



Question 3 **(****+)**

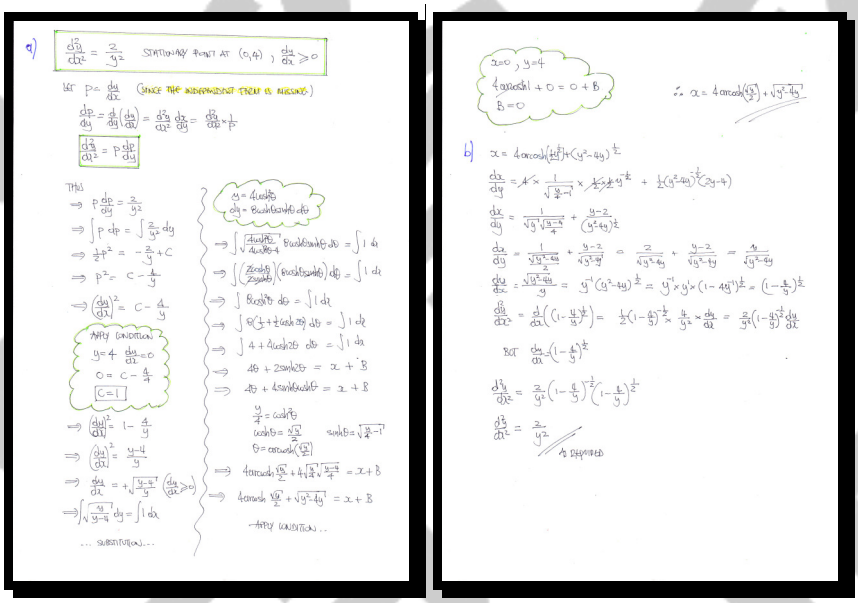
The curve C, has a stationary point at $(0,4)$ and satisfies the differential equation

$$\frac{d^2y}{dx^2}=\frac{2}{y^2},\ y\neq 0.$$

a) Given further that $\frac{dy}{dx}\geq 0$ along C, determine a simplified expression for the Cartesian equation of C, giving the answer in the form $x=f(y)$.

b) Verify by differentiation the answer to part **(a)**.

$$\boxed{x=4\operatorname{arcosh}\left(\frac{1}{2}\sqrt{y}\right)+\sqrt{y^2-4y}}$$



Question 4 (**+)**

The curve C with Cartesian equation $f(x,y)=0$, satisfies the differential equation

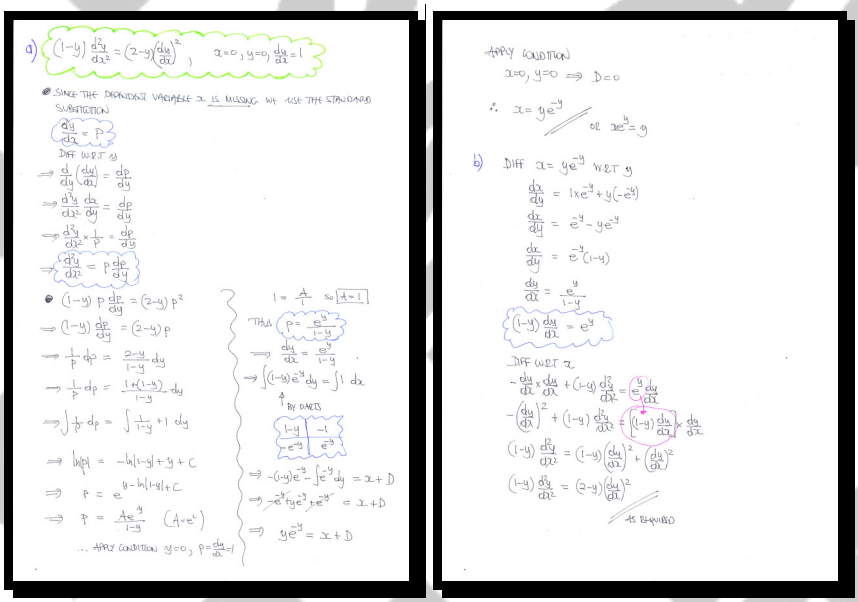
$$(1-y)\,y''=(2-y)(y')^2.$$

It is further given that $y(0)=0$ and $y'(0)=1$

a) Determine a simplified expression for the Cartesian equation of C.

b) Verify by differentiation the answer to part **(a)**.

$$\boxed{x=ye^{-y}}$$



Question 5 (*****)

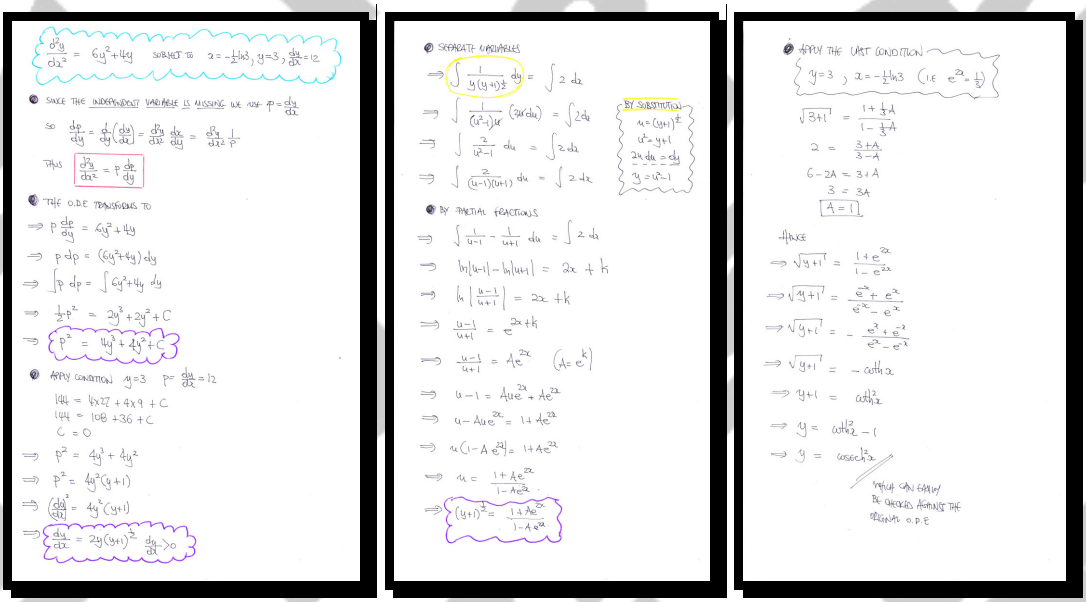
The curve with equation $y = f(x)$ satisfies the differential equation

$$\frac{d^2y}{dx^2} = 6y^2 + 4y, \quad \frac{dy}{dx} \geq 0.$$

If $y = 3$, $\frac{dy}{dx} = 12$ at $x = -\frac{1}{2}\ln 3$, solve the differential equation to show that

$$y = \operatorname{cosech}^2 x.$$

proof



Question 6 (*****)

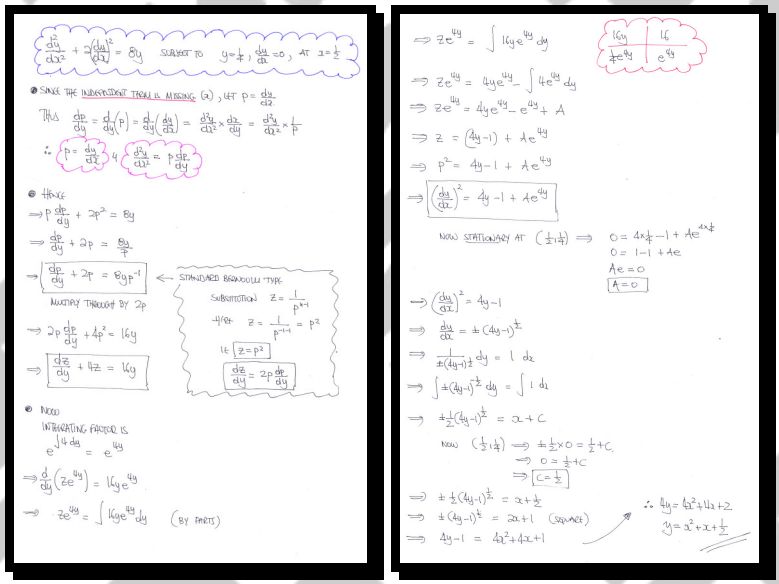
The curve with equation $y = f(x)$ satisfies the differential equation

$$\frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^2 = 8y .$$

Given further that the curve has a stationary point at $\left(\frac{1}{2}, \frac{1}{4}\right)$, solve the differential equation to show that

$$y = x^2 + x + \frac{1}{2} .$$

proof



Question 7 (*****)

The curve C, has gradient 1 at the origin and satisfies the differential relationship

$$\frac{d^2y}{dx^2}\sqrt{1-2y}=\frac{dy}{dx}(3y-2), \quad y<\frac{1}{2}.$$

Find an equation for C, giving the answer in the form $y=f(x)$.

$$y=\frac{\sin x}{1+\sin x}=(\sec x-\tan x)\tan x$$

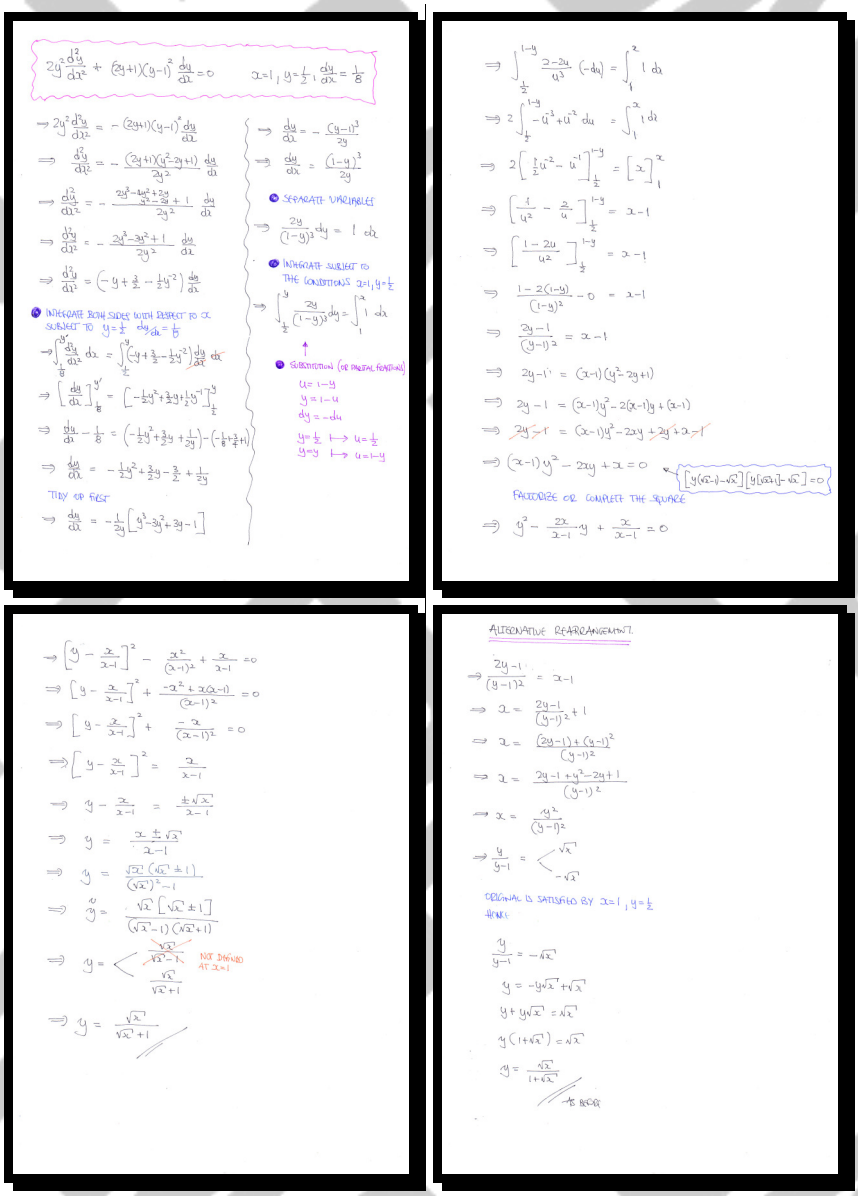
Question 8 (*****)

The curve C, has gradient $\frac{1}{8}$ at the point with coordinates $\left(1,\frac{1}{2}\right)$ and further satisfies the differential relationship

$$2y^2\frac{d^2y}{dx^2}+(2y+1)(y-1)^2\frac{dy}{dx}=0, \quad y\neq 0.$$

Find an equation for C, giving the answer in the form $y=f(x)$.

$$\boxed{y=\frac{\sqrt{x}}{1+\sqrt{x}}}$$



2^{ND} ORDER BY SUBSTITUTIONS

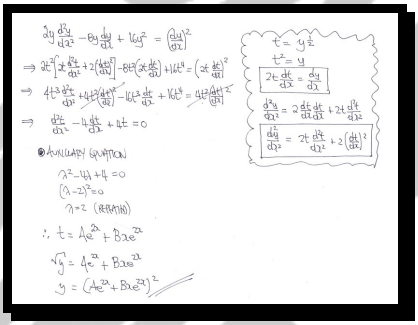
Question 1 (***)

$$2y\frac{d^2y}{dx^2}-8y\frac{dy}{dx}+16y^2=\left(\frac{dy}{dx}\right)^2,\ y\neq 0,$$

Find the general solution of the above differential equation by using the transformation equation $t=\sqrt{y}$.

Give the answer in the form $y=f(x)$.

$$\boxed{y=\left(A\mathrm{e}^{2x}+Bx\mathrm{e}^{2x}\right)^2}$$



Question 2 (***)

The differential equation

$$x\frac{d^2y}{dx^2}+2\frac{dy}{dx}=3x\,,\ x\neq 0\,,$$

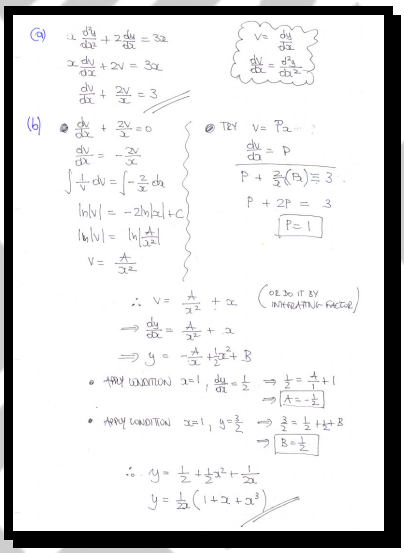
is to be solved subject to the boundary conditions $y=\frac{3}{2}$, $\frac{dy}{dx}=\frac{1}{2}$ at $x=1$.

a) Show that the substitution $v=\frac{dy}{dx}$, transforms the above differential equation into

$$\frac{dv}{dx}+\frac{2v}{x}=3\,.$$

b) Hence find the solution of the original differential equation, giving the answer in the form $y=f(x)$.

$$\boxed{y=\frac{1}{2}\left(x^2+\frac{1}{x}+1\right)}$$



Question 3 (***)

The curve C has equation $y=f(x)$ and satisfies the differential equation

$$x^2\frac{d^2y}{dx^2}-2x\frac{dy}{dx}-2y\left(2x^2-1\right)=3x^3\,\mathrm{e}^x\,,\quad x\neq 0$$

is to be solved subject to the boundary conditions $y=\frac{3}{2}$, $\frac{dy}{dx}=\frac{1}{2}$ at $x=1$.

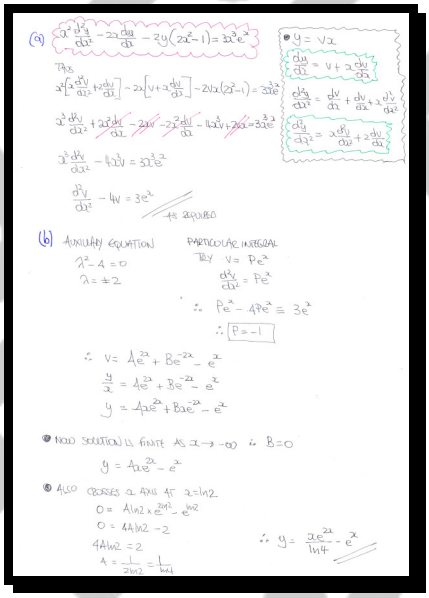
a) Show that the substitution $y=xv$, where v is a function of x transforms the above differential equation into

$$\frac{d^2v}{dx^2}-4v=3\mathrm{e}^x.$$

It is further given that C meets the x axis at $x=\ln 2$ and has a finite value for y as x gets infinitely negatively large.

b) Express the equation of C in the form $y=f(x)$.

$$\boxed{y=\frac{x\mathrm{e}^{2x}}{\ln 2}-\mathrm{e}^x}$$



Question 4 (*+)**

The differential equation

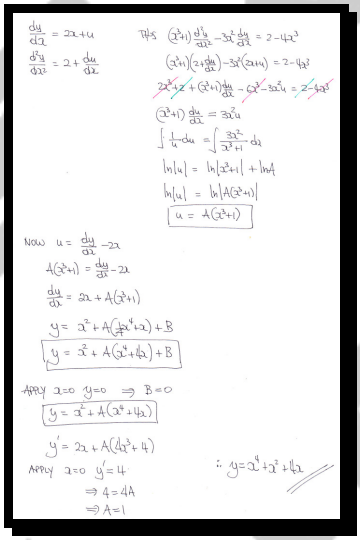
$$\left(x^3+1\right)\frac{d^2y}{dx^2}-3x^2\frac{dy}{dx}=2-4x^3,$$

is to be solved subject to the boundary conditions $y=0$, $\frac{dy}{dx}=4$ at $x=0$.

Use the substitution $u=\frac{dy}{dx}-2x$, where u is a function of x, to show that the solution of the above differential equation is

$$y=x^4+x^2+4x.$$

proof



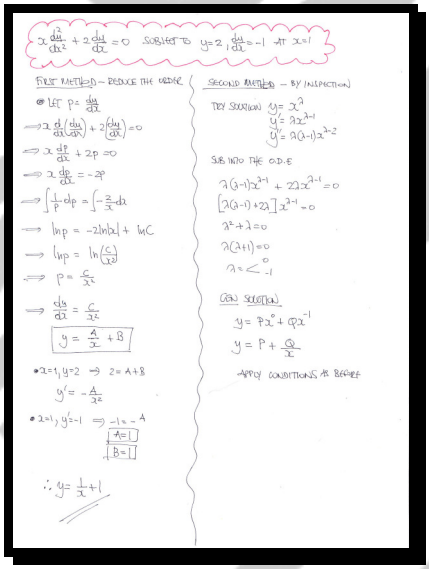
Question 5 (*+)**

Solve the differential equation

$$x\frac{d^2y}{dx^2}+2\frac{dy}{dx}=0,$$

subject to the boundary conditions $y=2$, $\frac{dy}{dx}=-1$ at $x=1$.

$$\boxed{y=\frac{2\mathrm{e}^{2x}}{\mathrm{e}^{2x}+x}}$$

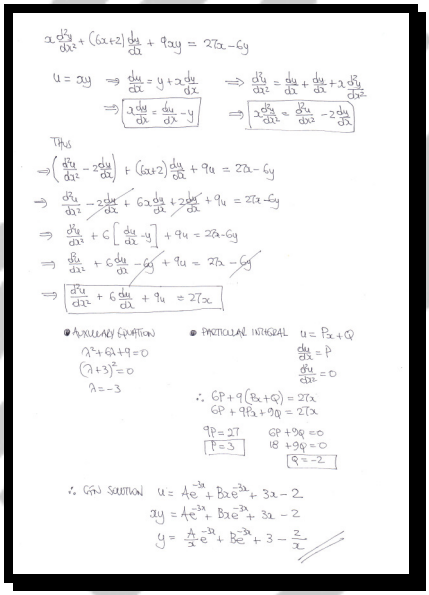


Question 6 (*+)**

$$x\frac{d^2y}{dx^2}+(6x+2)\frac{dy}{dx}+9xy=27x-6y\,.$$

Use the substitution $u=xy$, where u is a function of x, to find a general solution of the above differential equation.

$$\boxed{y=\frac{A}{x}\mathrm{e}^{-3x}+B\mathrm{e}^{-3x}+3-\frac{2}{x}}$$



Question 7 **(***+)**

$$\frac{d^2y}{dx^2}-2\frac{dy}{dx}\tan x-y\sec^4 x=0.$$

The above differential equation is to be solved by a substitution.

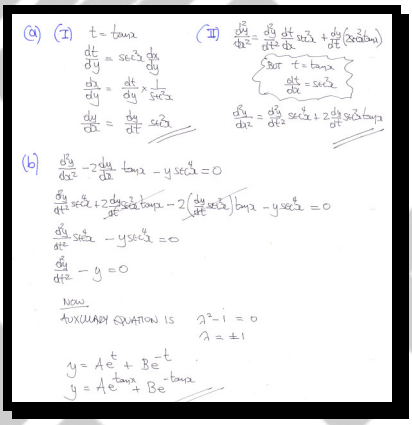
a) If $t=\tan x$ show that …

i. … $\dfrac{dy}{dx}=\dfrac{dy}{dt}\sec^2 x$

ii. … $\dfrac{d^2y}{dx^2}=\dfrac{d^2y}{dt^2}\sec^4 x+2\dfrac{dy}{dt}\sec^2 x\tan x$

b) Use the results obtained in part **(a)** to find a general solution of the differential equation in the form $y=f\left(x\right)$.

$$\boxed{y=A\mathrm{e}^{\tan x}+B\mathrm{e}^{-\tan x}}$$



Question 8 **(***+)**

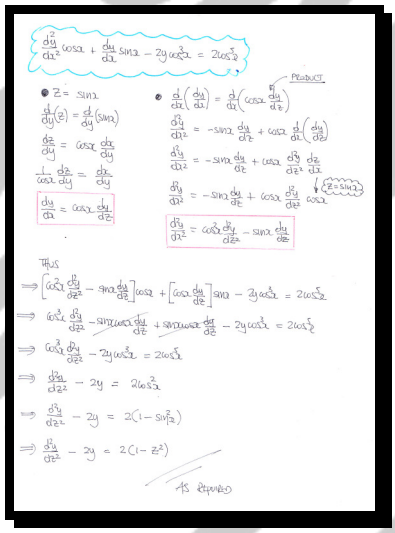
Show clearly that the substitution $z = \sin x$, transforms the differential equation

$$\frac{d^2y}{dx^2}\cos x + \frac{dy}{dx}\sin x - 2y\cos^3 x = 2\cos^5 x,$$

into the differential equation

$$\frac{d^2y}{dz^2} - 2y = 2\left(1 - z^2\right)$$

proof



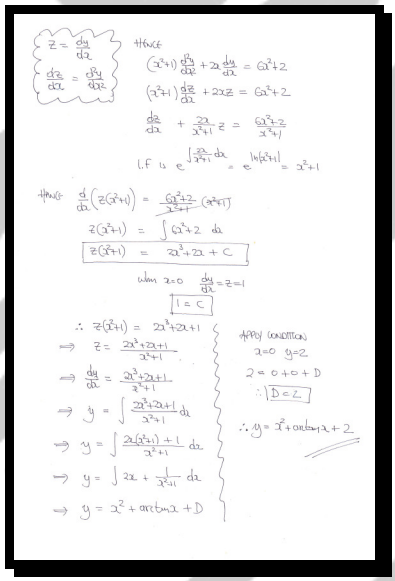
Question 9 (*+)**

By using the substitution $z = \dfrac{dy}{dx}$, or otherwise, solve the differential equation

$$\left(x^2+1\right)\frac{d^2y}{dx^2}+2x\frac{dy}{dx}=6x^2+2,$$

subject to the conditions $x=0$, $y=2$, $\dfrac{dy}{dx}=1$

$$\boxed{y=x^2+2+\arctan x}$$



Question 10 (****)

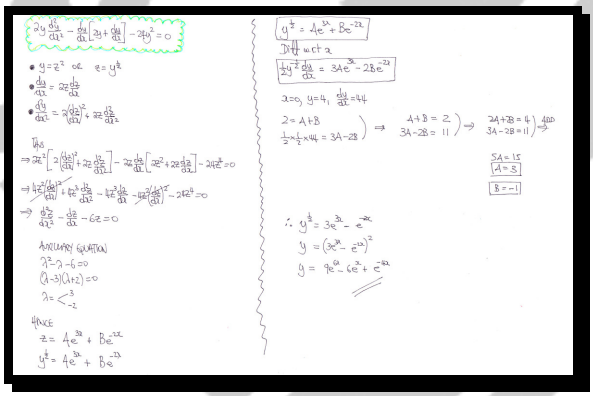
Use the substitution $z=\sqrt{y}$, where $y=f(x)$, to solve the differential equation

$$\frac{d^2y}{dx^2}+\frac{1}{y}\left(\frac{dy}{dx}\right)^2-5\frac{dy}{dx}+2y=0,$$

subject to the boundary conditions $y=4$, $\frac{dy}{dx}=44$ at $x=0$.

Give the answer in the form $y=f(x)$.

$$\boxed{y=9\mathrm{e}^{6x}-6\mathrm{e}^{x}+\mathrm{e}^{-4x}}$$



Question 11 (****)

$$2x\frac{d^2y}{dx^2}+\left(1-3x^{\frac{1}{2}}\right)\frac{dy}{dx}+y=0.$$

The above differential equation is to be solved by a substitution.

a) Given that $y=f\left(x\right)$ and $t=x^{\frac{1}{2}}$, show clearly that …

i. … $\dfrac{dy}{dx}=\dfrac{1}{2t}\dfrac{dy}{dt}$.

ii. … $\dfrac{d^2y}{dx^2}=\dfrac{1}{4t^2}\dfrac{d^2y}{dt^2}-\dfrac{1}{4t^3}\dfrac{dy}{dt}$.

b) Hence show further that the differential equation

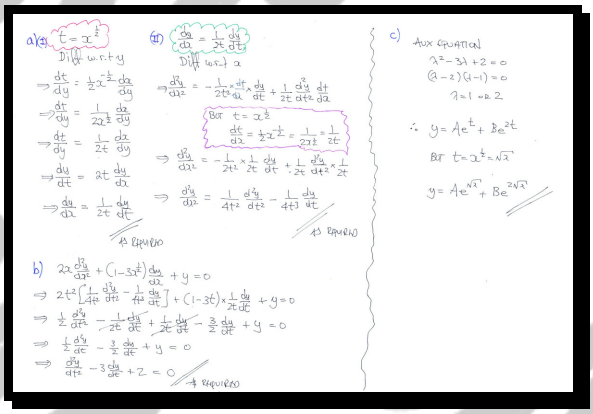
$$2x\frac{d^2y}{dx^2}+\left(1-3x^{\frac{1}{2}}\right)\frac{dy}{dx}+y=0,$$

can be transformed to the differential equation

$$\frac{d^2y}{dt^2}-3\frac{dy}{dt}+2y=0.$$

c) Find a general solution of the **original** differential equation, giving the answer in the form $y=f\left(x\right)$.

$$\boxed{y=A\mathrm{e}^{\sqrt{x}}+B\mathrm{e}^{2\sqrt{x}}}$$



Question 12 (****)

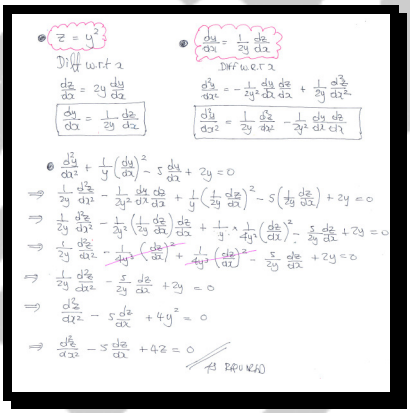
Show clearly that the substitution $z = y^2$, where $y = f(x)$, transforms the differential equation

$$\frac{d^2y}{dx^2} + \frac{1}{y}\left(\frac{dy}{dx}\right)^2 - 5\frac{dy}{dx} + 2y = 0,$$

into the differential equation

$$\frac{d^2z}{dx^2} - 5\frac{dz}{dx} + 4z = 0$$

proof



Question 13 (****)

Given that if $x = t^{\frac{1}{2}}$, where $y = f(x)$, show clearly that

a) $\dfrac{dy}{dx} = 2t^{\frac{1}{2}} \dfrac{dy}{dt}$.

b) $\dfrac{d^2y}{dx^2} = 4t\dfrac{d^2y}{dt^2} + 2\dfrac{dy}{dt}$.

The following differential equation is to be solved

$$x\frac{d^2y}{dx^2} - \left(8x^2+1\right)\frac{dy}{dx} + 12x^3y = 12x^5,$$

subject to the boundary conditions $y = \dfrac{10}{3}$, $\dfrac{d^2y}{dx^2} = 10$ at $x = 0$.

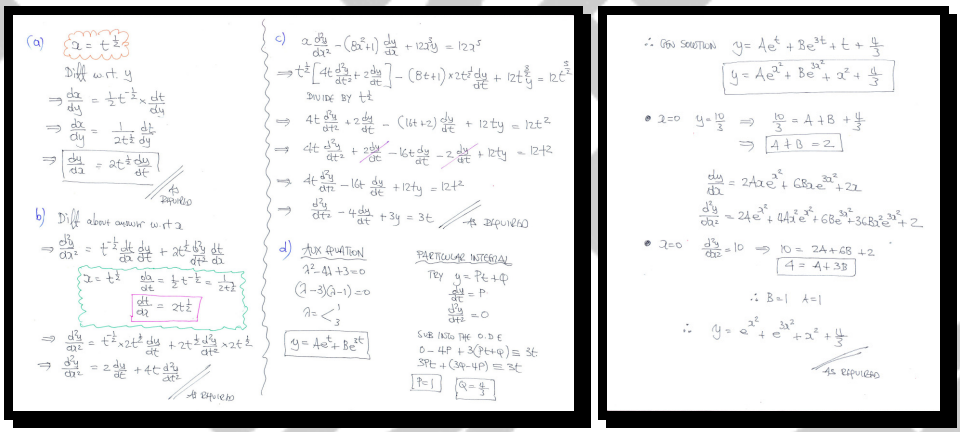
c) Show further that the substitution $x = t^{\frac{1}{2}}$, where $y = f(x)$, transforms the above differential equation into the differential equation

$$\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 3y = 3t.$$

d) Show that a solution of the **original** differential equation is

$$y = \mathrm{e}^{3x^2} + \mathrm{e}^{x^2} + x^2 + \frac{4}{3}.$$

proof



Question 14 (****)

$$\frac{d^2y}{dx^2}-2\frac{dy}{dx}\cot x+2y\operatorname{cosec}^2 x=2\cos x-2\cos^3 x\,.$$

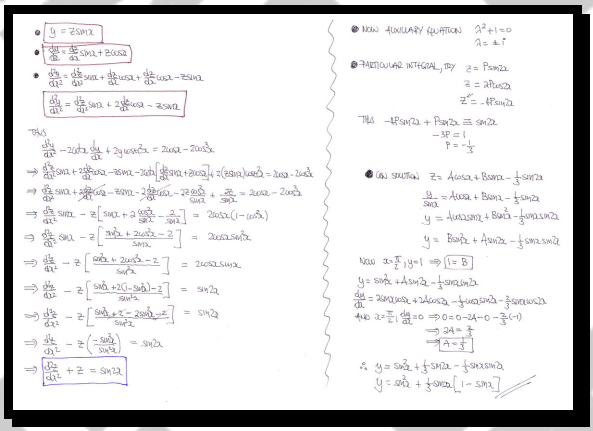
Use the substitution $y=z\sin x$, where z is a function of x, to solve the above differential equation subject to the boundary conditions $y=1$, $\dfrac{dy}{dx}=0$ at $x=\dfrac{\pi}{2}$.

Give the answer in the form

$$y=a\sin^2 x+b(1-\sin x)\sin 2x\,,$$

where a and b are constants to be found.

$$\boxed{a=1}\,,\quad \boxed{b=\frac{1}{3}}$$

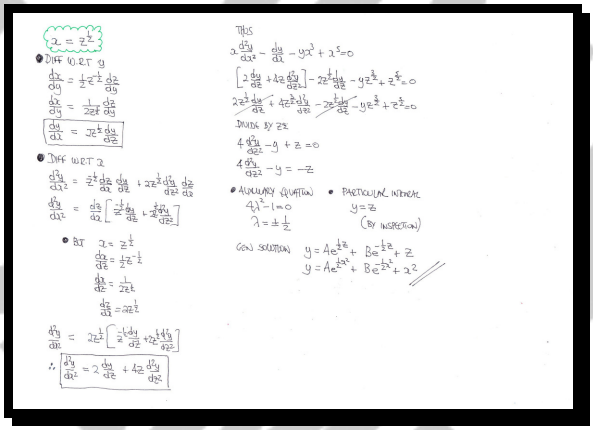


Question 15 (****)

$$x\frac{d^2y}{dx^2}-\frac{dy}{dx}-x^3y+x^5=0\,.$$

Use the substitution $x=z^{\frac{1}{2}}$, where $y=f\left(x\right)$, to find a general solution of the above differential equation.

$$\boxed{y=A\mathrm{e}^{\frac{1}{2}x^2}+B\mathrm{e}^{-\frac{1}{2}x^2}+x^2}$$



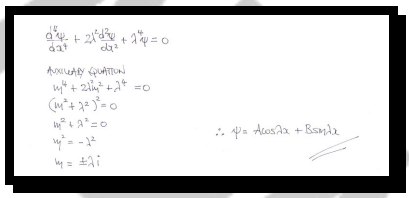
VARIOUS TYPES

Question 1 (+)**

Find the general solution of the following differential equation.

$$\frac{d^4\psi}{dx^4}+2\lambda\frac{d^2\psi}{dx^2}+\lambda^4\psi=0\,.$$

$$\boxed{\psi=A\cos\lambda x+B\sin\lambda x}$$



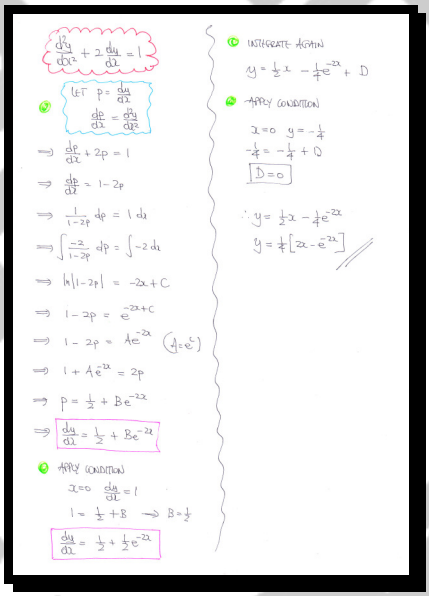
Question 2 (***)

Solve the differential equation

$$\frac{d^2y}{dx^2}+2\frac{dy}{dx}=1,$$

given that $y=-\frac{1}{4}$ and $\frac{dy}{dx}=1$ at $x=0$, giving the answer in the form $y=f(x)$.

$$\boxed{y=\frac{1}{2}\left[2x-\mathrm{e}^{-2x}\right]}$$



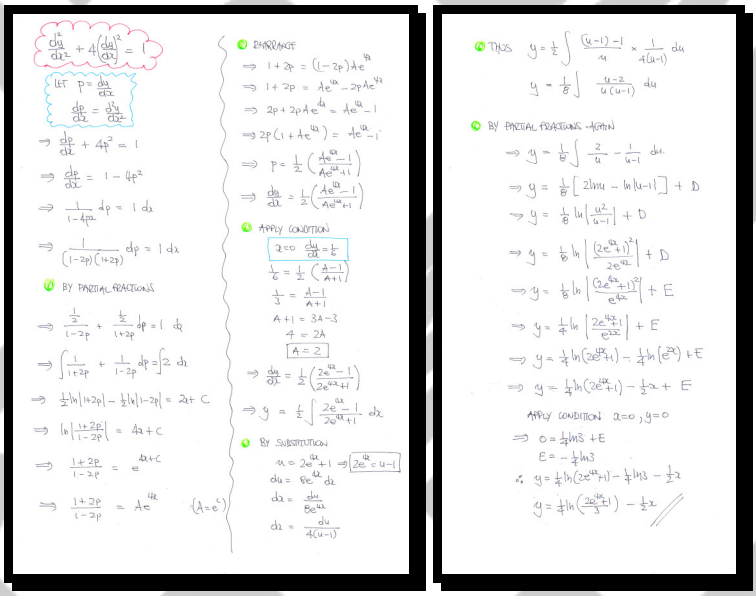
Question 3 (*+)**

Solve the differential equation

$$\frac{d^2y}{dx^2}+4\left(\frac{dy}{dx}\right)^2=1,$$

given that $y=0$ and $\frac{dy}{dx}=\frac{1}{6}$ at $x=0$, giving the answer in the form $y=f(x)$.

$$\boxed{y=\frac{1}{4}\ln\left[\frac{1+2\mathrm{e}^{4x}}{3}\right]-\frac{1}{2}x}$$



Question 4 (****)

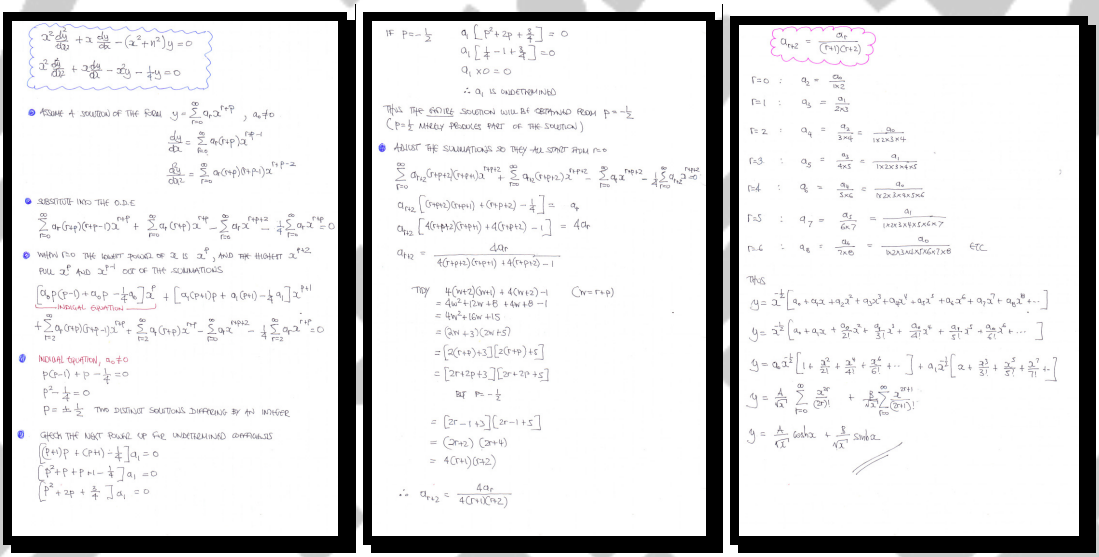
$$x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}-\left(x^2+n^2\right)y=0.$$

The above differential equation is known as modified Bessel's Equation.

Use the Frobenius method to show that the general solution of this differential equation, for $n=\frac{1}{2}$, is

$$y=x^{-\frac{1}{2}}\left[A\cosh x+B\sinh x\right].$$

proof



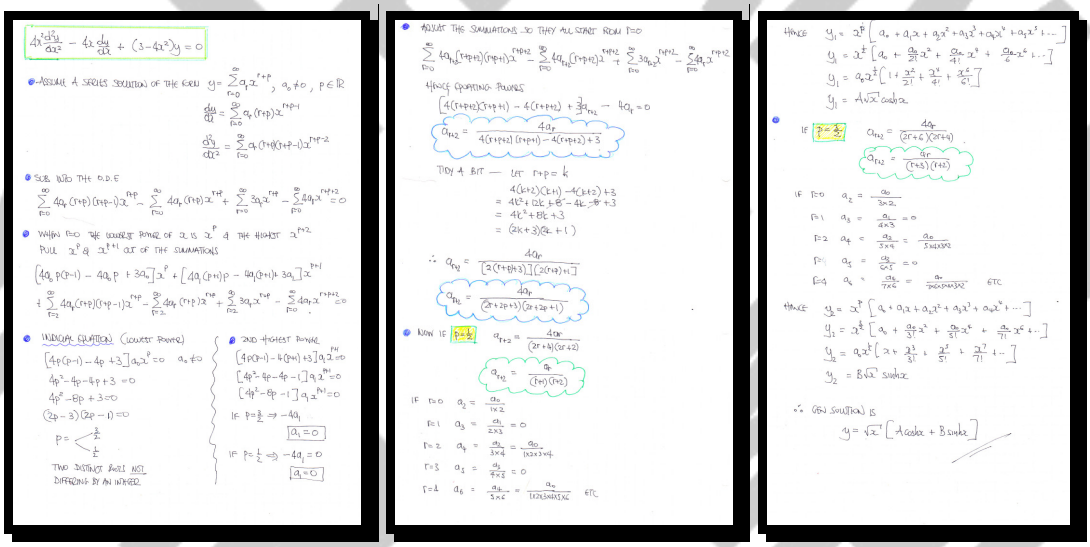
Question 5 (****)

Use the Frobenius method to find a general solution, as an infinite series, for the following differential equation

$$4x^2\frac{d^2y}{dx^2}-4x\frac{dy}{dx}+\left(3-4x^2\right)y=0\,.$$

Give the final answer in terms of elementary functions.

$$\boxed{y=\sqrt{x}\left(A\cosh x+B\sinh x\right)}$$



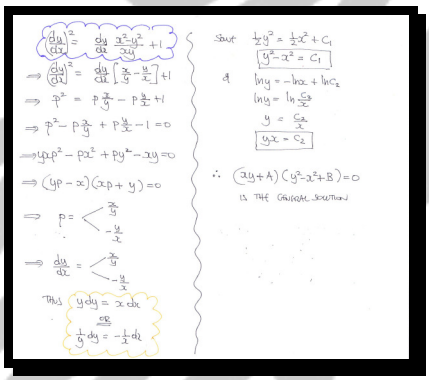
Question 6 (**+)**

By writing $\dfrac{dy}{dx}=p$ and seeking a suitable factorization find a general solution for the non linear differential equation

$$\left(\frac{dy}{dx}\right)^2=\frac{dy}{dx}\left(\frac{x^2-y^2}{xy}\right)+1.$$

Give the solution in the form $F(x,y)G(x,y)=0$.

$$\boxed{(xy+A)\left(x^2-y^2+B\right)=0}$$



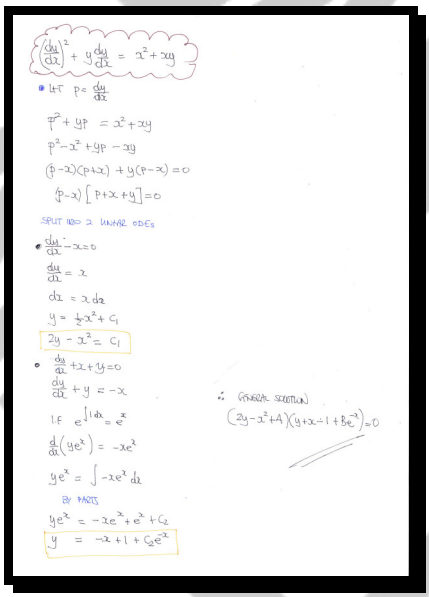
Question 7 (**+)**

By writing $\frac{dy}{dx} = p$ and seeking a suitable factorization find a general solution for the non linear differential equation

$$\left(\frac{dy}{dx}\right)^2 + y\frac{dy}{dx} = x^2 + xy\,.$$

Give the solution in the form $F(x,y)G(x,y) = 0$.

$$\boxed{\left(2y - x^2 + A\right)\left(x + y - 1 + B\mathrm{e}^{-x}\right) = 0}$$



Question 8 (*****)

A curve C is described implicitly by the equation

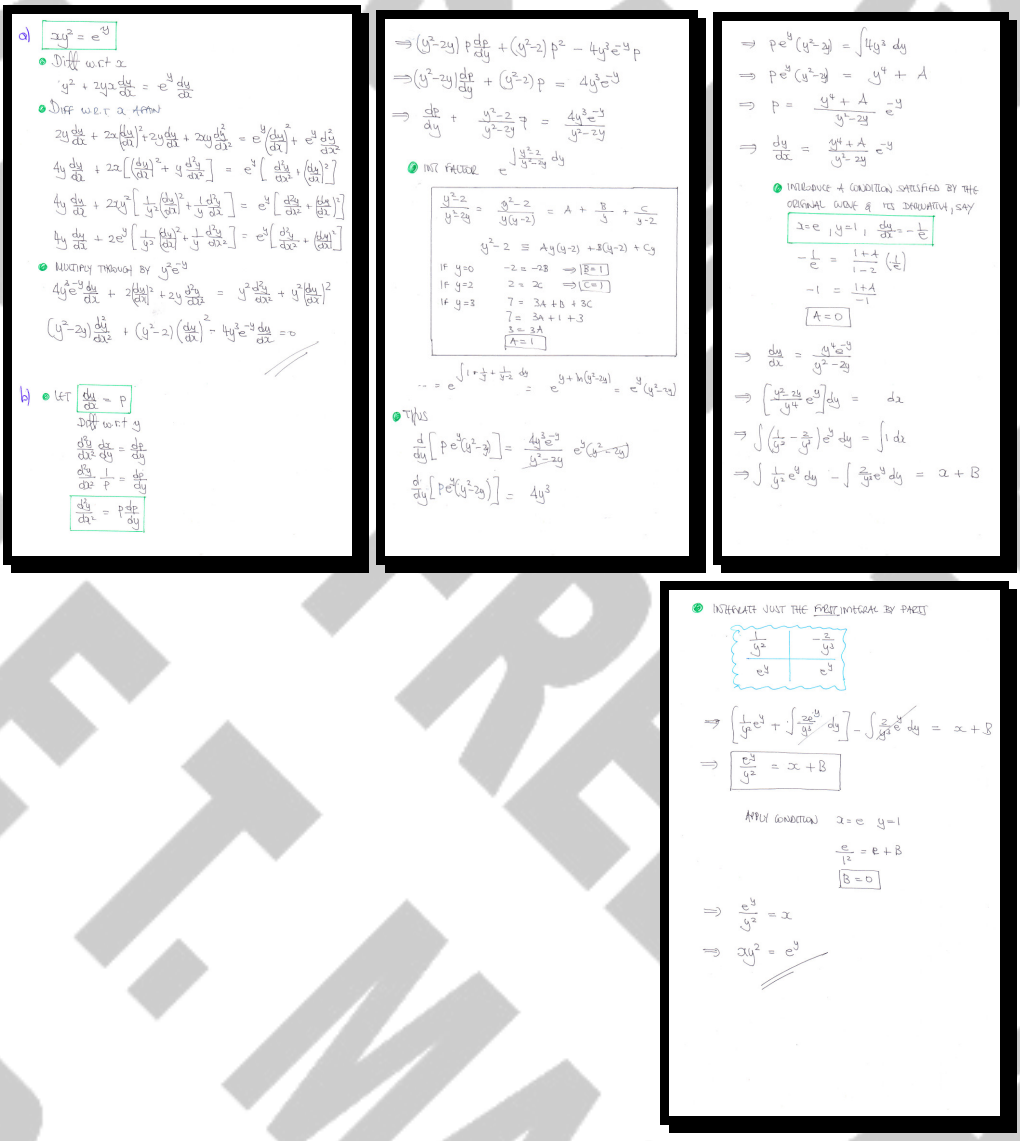
$$xy^2 = \mathrm{e}^y .$$

a) Show, by a detailed method, that

$$\left(y^2-2y\right)\frac{d^2y}{dx^2}+\left(y^2-2\right)\left(\frac{dy}{dx}\right)^2-4y^3\frac{dy}{dx}\mathrm{e}^{-y}=0 .$$

b) Use an analytical method, with suitable boundary conditions, to obtain the equation of C by solving the above differential equation.

proof

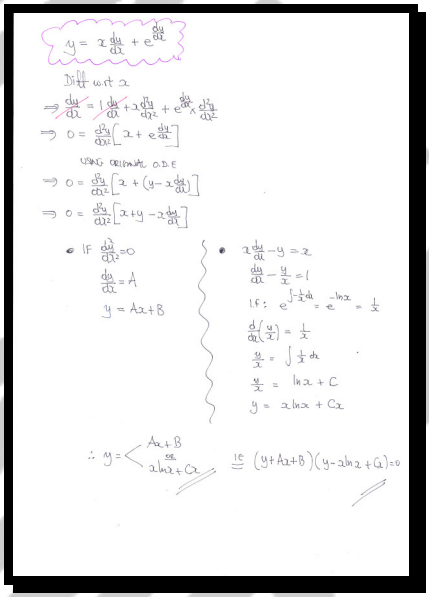


Question 9 (*****)

Find a general solution of the following differential equation.

$$y = x\frac{dy}{dx} + \mathrm{e}^{\frac{dy}{dx}}.$$

$$\boxed{\left(y + Ax + B\right)\left(y - x\ln x + Cx\right) = 0}$$



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